Multilungies: Multimodal lung transplant

Project description:

Literature review:

Todos

2025.05.27

- Lucy: Send a promising object w ScVI embeddings of single cell RNA seqs
- Dinesh: Patient ID and dates aligned for Radiomics data?
- Lucy: UNI embeddings for H&E → Dinesh, Initial clustering → Dinesh
- Make sure Dinesh has all simple modalities
- Dinesh: please check which modalities occur close together in timespace, does using 3(all) or 2 (h&e and CT scan) alone make sense?
- Dinesh: Storage organization in File Server
- data in “/lustre/groups/aih/dinesh.haridoss/”

—----------

The metadata (~1500 rows) given to me and the embedding (~800 after overlap) dates don’t match for h&e.

For any df that has more than one set of dates, please write here the file name and actual dates. 10metadata.csv has columns with dates, which is real and which is shifted, just to confirm.

scRNA directory? maybe you (Lucy) didn't get it yet?

2025.06.10

- Dinesh: group along scRNA into windows, check if we get better windows.
- Dinesh: also include 30 days into the analysis along with 100 and 365 days.
- Dinesh: set up file server, IT at helmholtz
- Lucy: give directories of scRNA, annotations of cell types or patch types, to Dinesh.
- Lucy: UMAP …

where do the datasets live:

scrna:
h&e:

what does the model do:

results:

on unimodal data:

on multimodal data:

Interpretable Multimodal Multiple Instance Learning for Lung Transplant Outcome Prediction

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1 Overview

We present a two-phase multimodal Multiple Instance Learning (MIL) framework for predicting outcomes in lung transplant patients. The framework integrates four distinct data modalities:

- **H&E histology (HE)** – high-dimensional histopathological imaging of biopsy tissue sections, represented as bags of patch-level embeddings (UNI features, 1024-dim)
- **Bronchoalveolar Lavage (BAL, scRNA-seq)** – single-cell transcriptomics from bronchoscopic lavage, represented as bags of cell-level scVI embeddings (10-dim)
- **Computed Tomography (CT)** – 3D radiological imaging, represented as bags of patch-level embeddings (1024-dim)
- **Clinical tabular data** – longitudinal patient measurements including spirometry, immunosuppression levels, and clinical laboratory values

The framework is designed for a longitudinal clinical setting in which each patient contributes multiple biopsy-anchored timepoints across the transplant follow-up period. At each timepoint, one or more modalities may be available, and the model must handle arbitrary missingness patterns gracefully.

Each modality is represented as a bag of instances, with imaging modalities carrying spatial location information. H&E patches include tile coordinates (tile_left, tile_top) indicating their position on the biopsy slide, enabling position-aware attention map visualization. CT instances carry 3D voxel coordinates (d_voxel, h_voxel, w_voxel) locating each patch within the scan volume.

abmil attention based multiple instance learning.

unimodal baselines

multimodal abmil

benchmarks

Prediction Tasks

Pre-trained modality encoders are either frozen (projection heads) or fine-tuned (backbones). A fusion module combines all available modality representations. We evaluate six fusion strategies: early, late, middle, cross-attention, cross-modal slot, and iterative cross-modal fusion.

The model addresses three clinical prediction tasks:

- Binary classification: Rejection vs. non-rejection under extreme class imbalance
- Survival prediction: Time to patient death
- CLAD prediction: Time to Chronic Lung Allograft Dysfunction occurrence

Model performance is evaluated using balanced accuracy (bAcc) and AUROC for classification tasks, and Concordance Index (C-index) for survival analysis.

Prediction Tasks

The model addresses three clinical prediction tasks: -

Binary classification (cls): Acute Cellular Rejection (ACR) vs. non-rejection under extreme class imbalance. Loss: weighted hinge.

Survival prediction (surv): Time to next ACR event using gap-time formulation. Loss: Cox-Breslow partial likelihood.

Multi-task alternating (both_alt): Per-epoch alternation between cls and surv objectives. Model selection on 0.5·BAcc + 0.5·C-index.

2 Dataset Preprocessing

2.1 Raw Data Sources

The dataset is assembled from four raw data sources spanning two imaging modalities and two molecular/clinical data types. Table 1 summarizes the data sources:

Modality	File Type	Patient ID Column	Time Column
BAL (scRNA-seq)	.h5ad	record_id	date_from_id
H&E histology	.h5ad	record_id	biopsy_date
CT scan	.h5ad	Patient	date_of_CT
Clinical	.csv	record_id	spiro_date

Instance-level embeddings for imaging modalities are stored in the .obsm key: X_scVI (BAL) or X (H&E, CT). Per-instance cluster and cell-type labels are in .obs columns: resolution_v2 (BAL), subcluster_renamed (H&E), and leiden (CT).

Spatial metadata: H&E patches include tile_left and tile_top for pixel position on the slide. CT patches include d_voxel, h_voxel, and w_voxel for 3D voxel location in the scan volume.

2.2 Dataset Statistics and Modality Coverage

The cohort consists of longitudinal biopsy-anchored timepoints from lung transplant patients. Each timepoint may be associated with a rejection label (derived from ACR biopsy grade), CLAD occurrence, and patient death status.

The largest modality group is Clinical-only (257 patients), reflecting that clinical measurements are most frequently collected. A total of 60 patients have all four modalities available simultaneously.

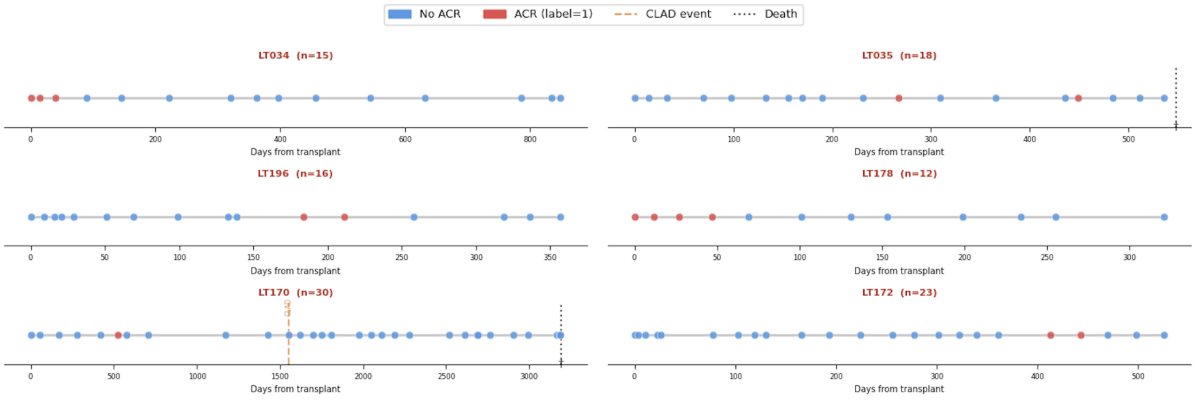


Figure 1 visualizes sample temporal distribution: the x-axis shows days from transplant, with each point representing a multimodal sample (containing at least one modality), colored by ACR status, CLAD occurrence, and death.

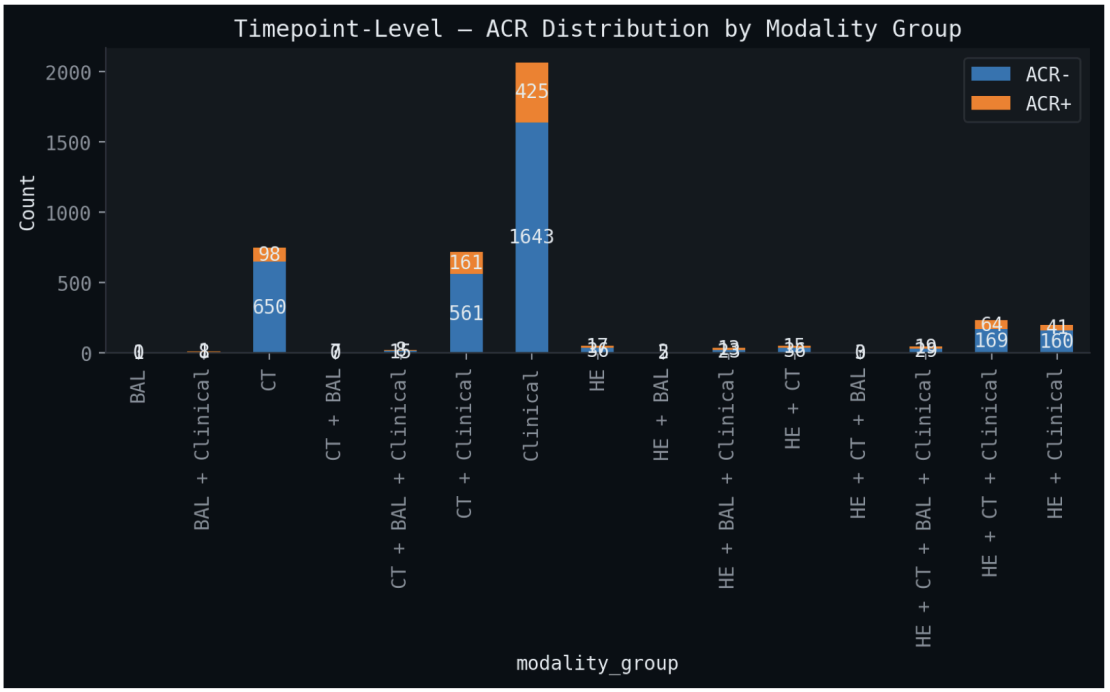


Figure 2 shows the distribution of ACR-positive and ACR-negative timepoints for each modality combination. The dataset exhibits severe class imbalance: rejection events are a minority across all modality groups. Clinical-only timepoints are most frequent (1,643 ACR−, 425 ACR+), while the HE+CT+Clinical group contributes 169 ACR− and 64 ACR+ timepoints.

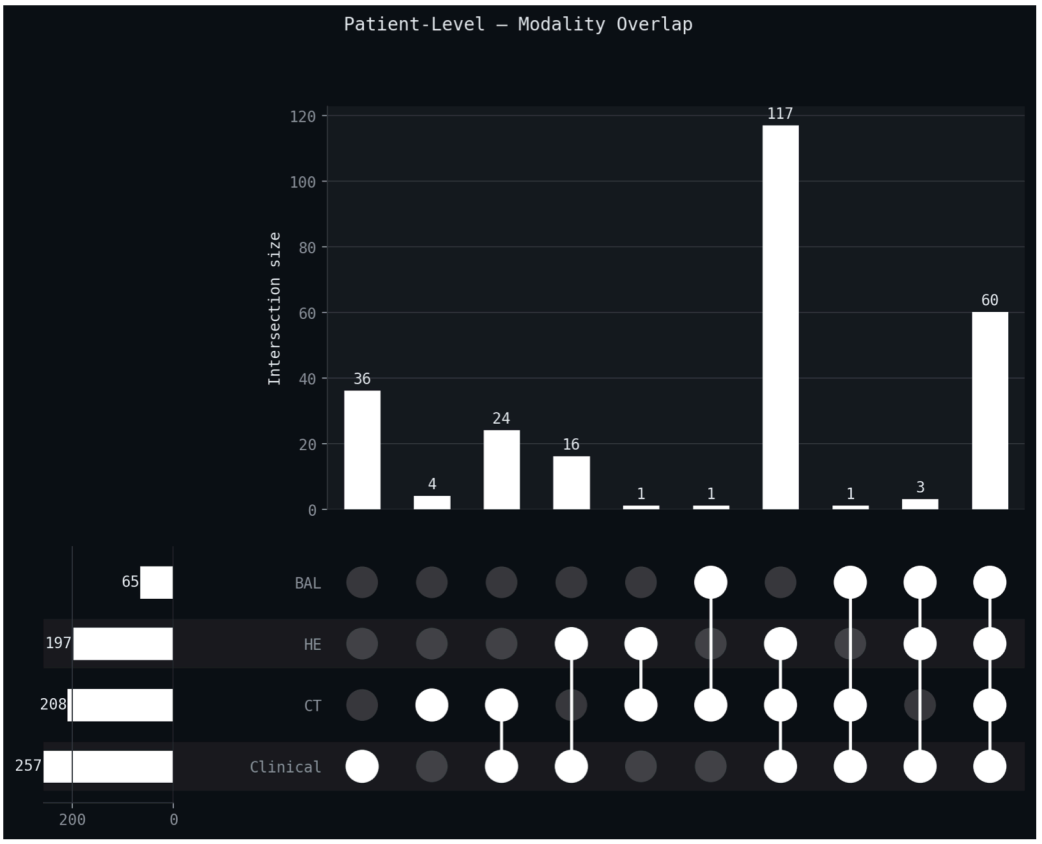
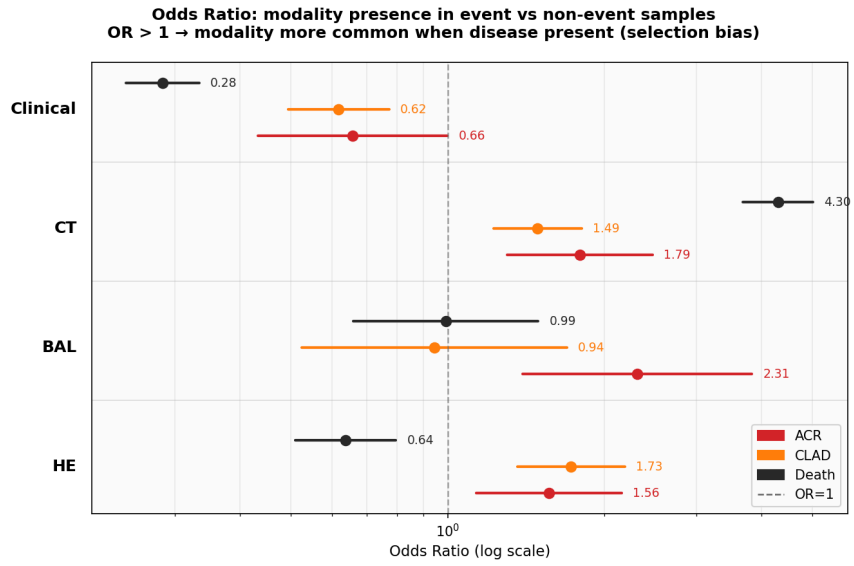
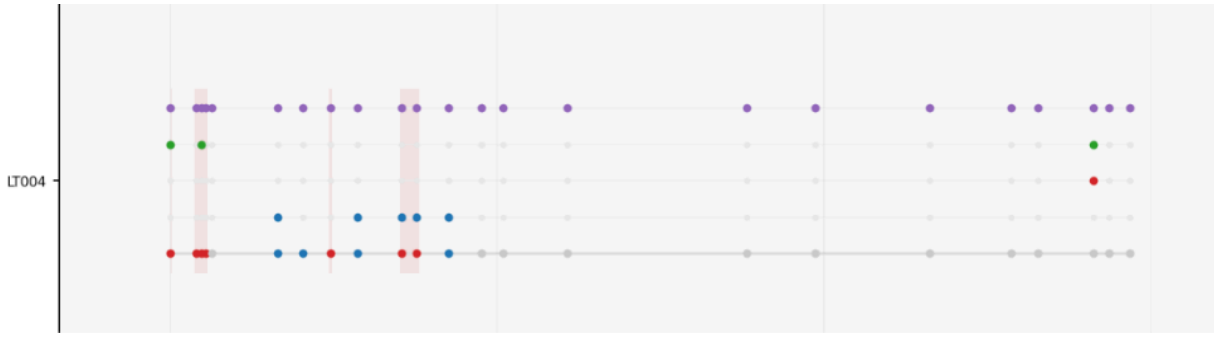


Figure 3 (UpSet plot) illustrates patient-level modality overlap. Each bar represents a modality combination, with Clinical-only being the largest single-modality group (257 patients) and the all-four-modality group containing 60 patients.





2.3 Temporal Alignment

All modalities are aligned to a common anchor event using a time-windowed nearest-neighbor strategy. The anchor modality is selected hierarchically: samples are first anchored and paired around the primary anchor, then successive modalities act as anchors until all modalities have been aligned.

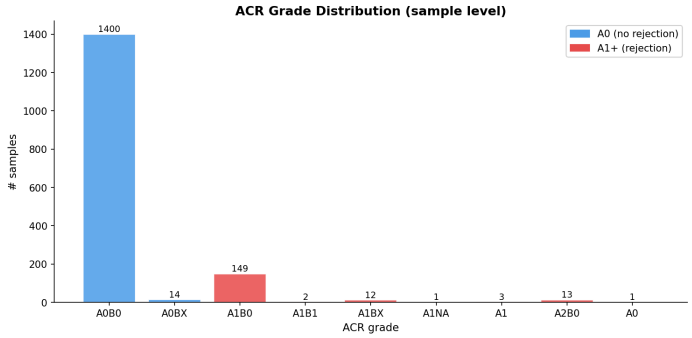
For each biopsy event, the algorithm searches for the nearest available measurement from every other modality within a ±45-day window (45 × 86,400 seconds). If multiple measurements from the same modality fall within this window, the temporally closest one is selected. A timepoint is included as long as at least one modality is present (min_modalities = 1).

The alignment procedure operates as follows:

1. Fix the anchor timestamp t_0 from the biopsy date
2. For each modality $m \in \{\text{BAL, CT, Clinical}\}$, collect all measurements from the same patient
3. Retain measurements where timestamp t_m satisfies $|t_m - t_0| \leq 45 \times 86{,}400$ seconds
4. Among retained measurements, select the one minimizing $|t_m - t_0|$ as the aligned record for modality m
5. If no measurement falls within the window, modality m is marked as absent for this timepoint

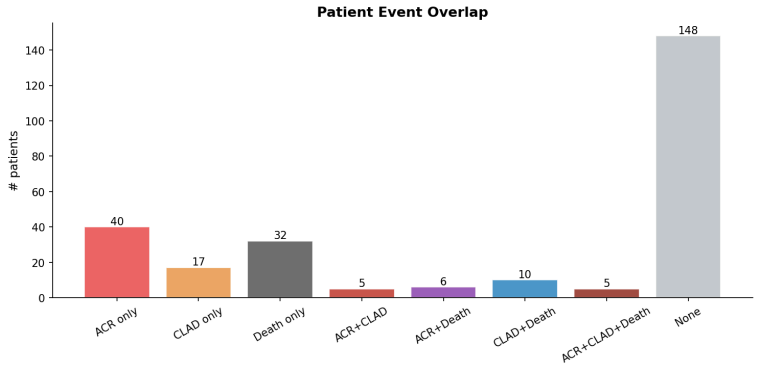
This procedure produces aligned multimodal samples. Each sample has one rejection label from the biopsy and zero or more co-occurring modality observations within the 45-day window. Patients without a biopsy on a given date can still contribute if their most recent biopsy label falls within the window.

2.4 Label Encoding

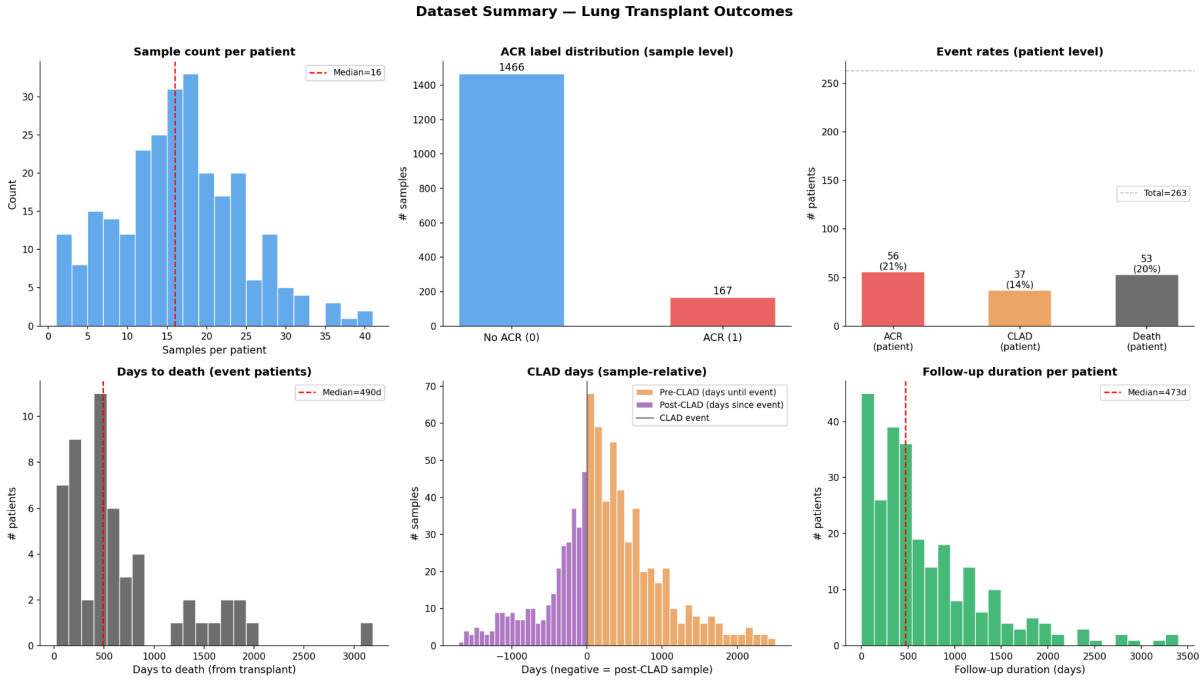


The binary rejection label is derived from the ACR biopsy grade:

y = 1 if ACR grade contains A1 or A2 (acute rejection)
y = 0 otherwise



2.5 Survival Outcomes (CLAD and Death)



Death Outcome

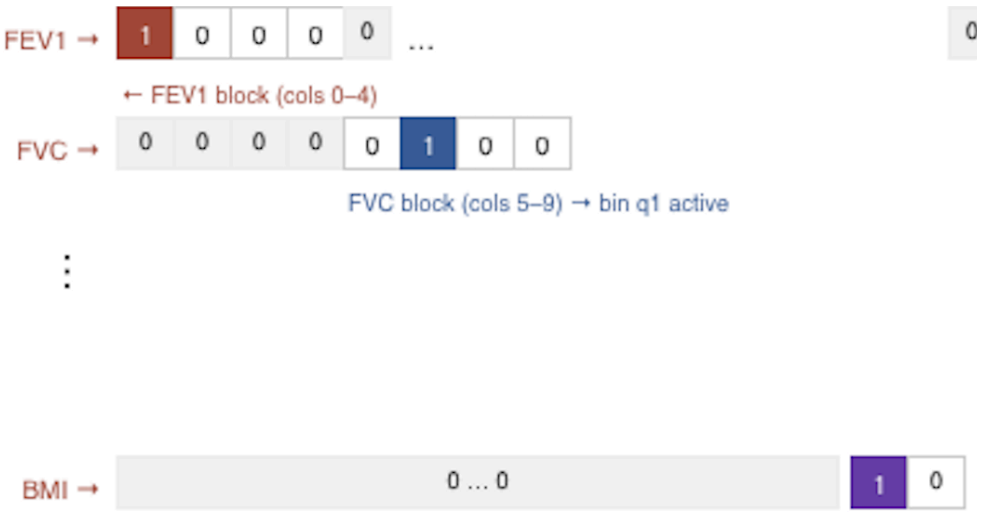
Death dates are recorded for deceased patients, enabling time-to-event analysis. Patients who have not died are right-censored at their last observation date. For deceased patients, all prior samples include the number of days from sample collection to death.

CLAD Outcome

For patients who developed Chronic Lung Allograft Dysfunction (CLAD), we record the number of days from sample collection to CLAD diagnosis. Post-CLAD samples are excluded from the analysis. Patients who remain CLAD-free are right-censored at their last follow-up.

Important: To prevent label leakage, outcome-predictive clinical features (FVC/FEV1 slopes, biopsy grade itself) are excluded from the clinical feature set used for prediction.

2.6 Clinical Feature Tokenization



Raw clinical tabular features (F = 102 features after label-leakage exclusion) are discretized by a ClinicalFeatureTokenizer:

Continuous Features

Each continuous feature f is discretized into nq = 4 quantile bins fitted on the population. Bin boundaries $b_0^{(f)} < b_1^{(f)} < ... < b\square q^{(f)}$ correspond to the 0, 1/nq, ..., 1 quantiles of observed values. A dedicated NaN token is reserved per feature for missing values.

Categorical/Binary Features

Features with ≤10 unique values are treated as categorical. Each unique category receives a distinct token, plus one NaN token for missing values.

Block-Diagonal One-Hot Representation

At precompute time, each sample's tokenized clinical vector $t \in \mathbb{Z}^f$ (one token ID per feature) is converted into a 2D bag matrix $X_Clin \in \{0,1\}^{(F \times V_Clin)}$. The global vocabulary has V_Clin tokens, partitioned into F non-overlapping blocks of size (nq + 1) for continuous features or (|categories| + 1) for categorical features.

Row f of X_Clin is a one-hot vector with a single 1 at the position of token t_x. Because each feature occupies a disjoint block of columns, the matrix is block-diagonal and each row represents a valid token embedding. This treats clinical features as F instance tokens on equal footing with imaging patch tokens.

For the special case of nq = 4 bins with all F features continuous, the representation simplifies to $X_Clin \in \{0,1\}^{(F \times 4F)}$ with feature dimension 4F = 408, matching the MODALITY_REGISTRY entry ('clinical_onehot', 408).

2.7 Cluster Count Tokenization

For each imaging modality m ∈ {HE, BAL, CT} with K□ cell or patch clusters, a per-bag cluster abundance vector c□ ∈ ℝ^K□ (CLR-normalized counts) is precomputed from the per-instance cluster labels. A ClusterCountTokenizer discretizes each cluster's abundance into nq = 4 quantile bins.

Each cluster receives its own separate set of quantile boundaries fitted on the population distribution for that specific cluster. The boundaries for cluster k are computed from the CLR-normalized count of cluster k across all patients. This per-cluster fitting ensures that bin thresholds are meaningful for each cell type independently, regardless of how common or rare that cell type is.

The token ID vector is t□ ∈ ℤ^K□, where entry k encodes the quantile bin of cluster k's abundance relative to the population. The analogous block-diagonal one-hot matrix is X^count_m ∈ {0,1}^(K□×nqK□): row k has a 1 at column k·nq + b□, where b□ is the quantile bin of cluster k's count.

Pooling across rows yields the bag-level count representation:

c^onehot_m = (X^count_m)ᵀ 1_K□ ∈ {0,1}^(nqK□)

This produces a sparse vector with exactly K□ non-zero entries (one per cluster).

2.8 Precomputed .pt File Format

Each aligned timepoint is saved as a PyTorch .pt dictionary with the following top-level keys (after key remapping):

- inputs: Dictionary of modality tensors – HE_cells (N_HE, 1024), BAL_cells (N_BAL, 10), CT_cells (N_CT, 1024), Clinical (N_Clin, d_Clin)
- cluster_labels: Per-instance cluster assignment strings
- coords: Per-instance spatial coordinates
- bag_centroids: Dictionary of per-modality centroid matrices (K□, d□)
- clinical_onehot: X_Clin ∈ {0,1}^(F×4F)
- cluster_count_onehot: Dictionary of X^count_m matrices
- cluster_names: Dictionary of cluster name lists per modality
- clinical_feature_names: List of F feature name strings
- clinical_token_ids: Per-patient bin assignment indices
- clinical_vocab: List of bin-specific token label dictionaries (e.g., {'id': 3, 'label': 'FEV1_q3'})
- label, identifier, anchor_time, metadata, survival: Outcome and metadata fields

3 Data Representation

3.1 Modality Feature Spaces

Table 2 summarizes the modality registry. Each bag is represented as a set of instance-level feature vectors with specified dimensions and assigned roles in the teacher-student framework.

Modality	Key	Feature Dimension d	Role
H&E	HE_cells	1024	Teacher
Clinical	clinical_onehot	408	Teacher
CT	CT_cells	1024	Student
BAL	BAL_cells	10	Student

Clinical Representation: Clinical features are encoded as a 2D bag X_Clin ∈ ℝ^(F×F·nb), where F = 102 is the number of clinical features and nb = 4 is the number of quantization bins. Row f is a block-diagonal one-hot vector encoding the binned value of feature f. There is exactly one non-zero entry at column f·nb + bf, where bf is the bin index.

This representation treats each clinical feature as a separate token instance, enabling attention-based pooling over clinical features on equal footing with histological patch tokens.

4 Phase 1: Unimodal Pre-Training

4.1 Gated Attention MIL Encoder

Following Ilse et al. [1], each modality encoder maps a bag X ∈ ℝ^(N×d) (with N instances) to a scalar logit via three stages:

Instance Projection (Backbone)

$h_i = \mathrm{ReLU}(W_{bb} x_i),\ h_i \in \mathbb{R}^H,\ H = 256$

Gated Attention Pooling

$a_i = \exp(w^T[\tanh(Vh_i) \odot \sigma(Uh_i)]) / \Sigma_{\square} \exp(w^T[\tanh(Vh_{\square}) \odot \sigma(Uh_{\square})])$

where V, U ∈ ℝ^(H×H), w ∈ ℝ^H

Bag Representation and Classification

$r = \Sigma_i a_i h_i,\ \hat{y} = w_{head}^T r$

4.2 Phase 1 Loss Functions

4.2.1 Task Loss (Weighted Hinge)

$\mathcal{L}_{task} = (1/B) \Sigma_b w_b \cdot \max(0, 1 - \tilde{y}_b \hat{y}_b)$

where $\tilde{y}_b = 2y_b - 1 \in \{-1, +1\}$ and w_b is a class weight that upweights the minority (rejection) class.

4.2.2 Bidirectional Patch Cross-Attention

For student–teacher pairs, we compute bidirectional patch-level cross-attention following the MCAT architecture [2]. Let Hs ∈ ℝ^(Ns×H) and Ht ∈ ℝ^(Nt×H) be the student and teacher patch features.

Student Attends Teacher:

$$CA_{s\to t} = \mathrm{softmax}(Q_s K_t^\top / \sqrt{H/nh})\, V_t$$

This produces enriched student features H̃s. The multi-head variant uses nh = 4 heads. After cross-attention, layer normalization and a feed-forward residual are applied.

Knowledge Distillation Signal: For knowledge distillation, the teacher attends to student embeddings. The teacher patches act as queries and the student patches act as keys and values. This produces cross-attention weights $\alpha_{t\to s} \in \mathbb{R}^{(N_t \times N_s)}$, where entry [j, i] measures how strongly teacher patch j attends to student patch i.

These attention weights are computed from the pre-softmax logits $Q_t K_s^\top / \sqrt{H/nh}$. The per-student-patch importance score is obtained by averaging over teacher patches:

$$\bar{\alpha}_{t\to s} = (1/N_t) \sum_j \alpha_{t\to s}[j, :] \in \mathbb{R}^{N_s}$$

This score reveals which student patches the teacher considers most informative, serving as a soft target for the student's own self-attention.

4.2.3 Attention Transfer Loss (KD)

Inspired by Zagoruyko & Komodakis [3], we align the student's self-attention distribution to the teacher-guided distribution over the top-K most attended student patches. Both α_{self} and $\bar{\alpha}_{t\to s}$ are pre-softmax attention logits. Softmax is applied inside the KL divergence to convert them to distributions:

$$\mathcal{L}_{KD} = \mathrm{KL}(\mathrm{softmax}(\alpha_{self}[K]/\tau) \,||\, \mathrm{softmax}(\bar{\alpha}_{t\to s}[K]/\tau))$$

where K is the index set of the top-K = 50 student patches by teacher score, and τ = 0.5 is a temperature parameter. Using pre-softmax logits allows the temperature τ to control the sharpness of the target distribution independently of the attention scores.

4.2.4 Cosine Representation Distillation (CRD)

To align the student's cross-attended bag representation r×s with the cached teacher summary rt:

$$\mathcal{L}_{CRD} = 1 - (r_{\times s} / ||r_{\times s}||) \cdot (r_t / ||r_t||)$$

This loss is adapted from Contrastive Representation Distillation [4].

4.2.5 Supervised Contrastive Loss (SupCon)

All modality representations z = g(r) (projected through a two-layer MLP projection head g: ℝ^H → ℝ^128) contribute to a Supervised Contrastive loss [5]:

$\mathcal{L}_{CLR} = -\Sigma_i w_i (1/|P(i)|) \Sigma_{□ \in P(i)} \log[\exp(z_i \cdot z_□/\tau c) / \Sigma_{□ \neq i} \exp(z_i \cdot z_□/\tau c)]$

with $\tau c = 0.07$ and positive set $P(i) = \{j : (\text{same stem} \wedge \text{different modality}) \vee (\text{same label} \wedge \text{different stem})\}$. Class weights w_i upweight the minority class.

An analogous summary CLR loss is computed on the normalized bag representations $\ell_2(r)$ directly, without a projection head.

4.2.6 Total Phase 1 Objective

$\mathcal{L}_1 = \mathcal{L}_{task} + \lambda_{cx} \mathcal{L}_{cross} + \lambda_{kd} \mathcal{L}_{KD} + \lambda_{crd} \mathcal{L}_{CRD} + \lambda_{clr} \mathcal{L}_{CLR} + \lambda_{sclr} \mathcal{L}_{sCLR}$

where $\mathcal{L}_{cross}$ is an auxiliary hinge loss computed from the cross-attended student representation. The default configuration uses task loss only, with all λ coefficients set to 0 unless explicitly enabled.

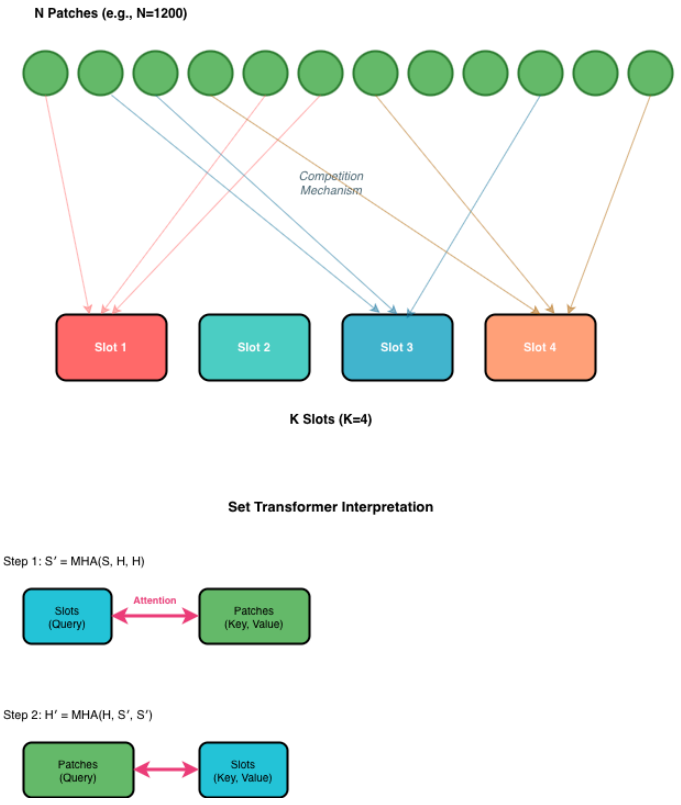
4.3 Phase 1 Hyperparameters

Table 3 summarizes the hyperparameters used during Phase 1 training:

Parameter	Symbol	Value
Hidden dimension	H	256
Dropout	p	0.4
Learning rate	η_1	1×10^{-5}
Weight decay	—	1×10^{-3}
Max epochs	—	400
Eval frequency	—	Every 25 epochs
Gradient accumulation	—	4 steps
Cross-attention heads	nh	4
Projection dimension	dz	128
KD temperature	τ	0.5
KD top-K	K	50
CLR temperature	τc	0.07
λ_cx	—	0.05
λ_kd	—	0.05
λ_crd	—	0.10
λ_clr	—	0.10
λ_sclr	—	0.10

5 Phase 2: Multimodal Fusion

5.1 Shared Building Blocks



5.1.1 Iterative Slot Attention

Slot Attention [6] compresses a variable-length patch sequence H ∈ ℝ^(N×H) into K fixed-size slot vectors S ∈ ℝ^(K×H), where K ≪ N. Each slot learns to represent a different subset of the input patches through a competition mechanism.

Set Transformer Interpretation: The core idea is related to the Induced Set Attention Block (ISAB) from the Set Transformer [7]. Two attention operations define the block:

First, the K slot vectors act as queries and the N patch features act as keys and values:
S′ = MHA(S, H, H)

Each slot gathers information from the patches it is most similar to.

Second, the patches can be refined by attending back to the updated slots:
H′ = MHA(H, S′, S′)

Here S is a small set of K vectors in ℝ^d, with K ≪ N. This bottleneck forces the slots to form compact summaries of the full patch sequence.

Slot Attention Procedure: Our implementation follows Locatello et al. [6] and uses a GRU-based iterative update. A slot-competition softmax is applied column-wise over patches so that each patch is drawn toward the slot it fits best.

Given patch features $H \in \mathbb{R}^{N \times H}$, slot count K, and iterations T:

Initialize $S^{(0)} \sim N(\mu, \text{softplus}(\sigma))$ from learned parameters

Compute keys and values once: $k = W_k\,\text{LN}(H)$, $v = W_v\,\text{LN}(H)$

For $t = 1, \ldots, T$:

(a) $q \leftarrow W_q\,\text{LN}(S^{(t-1)})$

(b) $L \leftarrow qk^{T}/\sqrt{H} \in \mathbb{R}^{K \times N}$

(c) $A \leftarrow \text{softmax_K}(L)$ (softmax over slots, not patches)

(d) $\tilde{A} \leftarrow A \odot (A1_N + \varepsilon)^{-1}$ (normalize each slot)

(e) $U \leftarrow \tilde{A}\,v$ (weighted mean of patch values per slot)

(f) $S^{(t)} \leftarrow \text{GRU}(U, S^{(t-1)})$

(g) $S^{(t)} \leftarrow \text{LN}(S^{(t)} + \text{FFN}(S^{(t)}))$

Return $S^{(T)} \in \mathbb{R}^{K \times H}$

Default settings: $K \in \{8, 16\}$, $T = 3$

5.1.2 Cross-Modal Transformer

Slot tokens from all present modalities are concatenated and refined by an L-layer Transformer encoder:

$$Z^{(l+1)} = \text{FFN}(\text{LN}(Z^{(l)}) + \text{MHA}(Z^{(l)}, Z^{(l)}, Z^{(l)}))$$

with $L = 2$ layers and $n_h = 4$ attention heads. The FFN uses a GELU activation with hidden dimension $2H$.

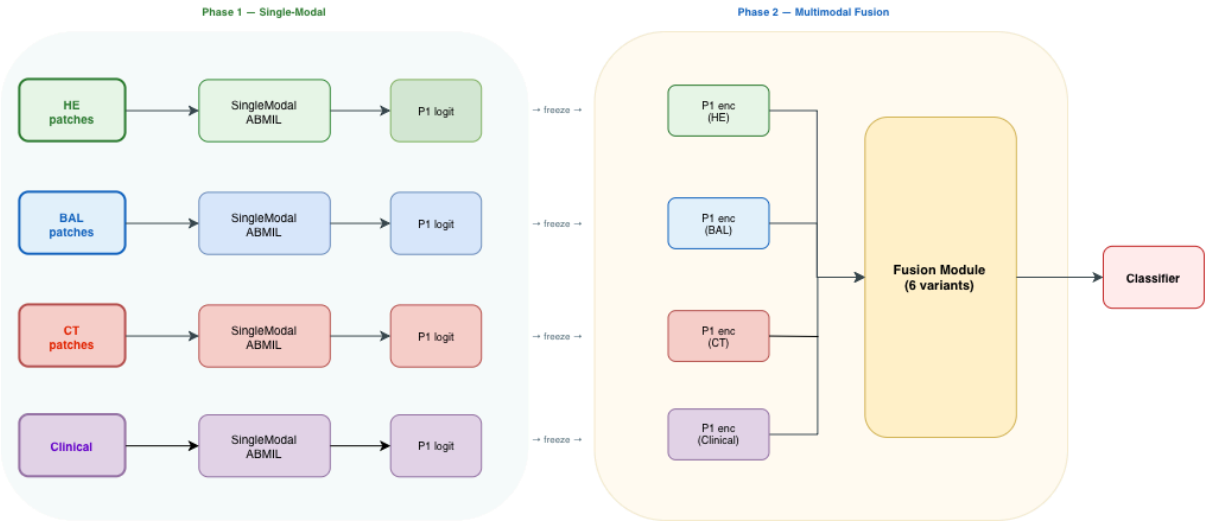
5.1.3 Final Pooling

Two pooling modes are supported after the cross-modal transformer:

ABMIL Pooling: Standard gated attention (Equation 4) applied to the slot token sequence.

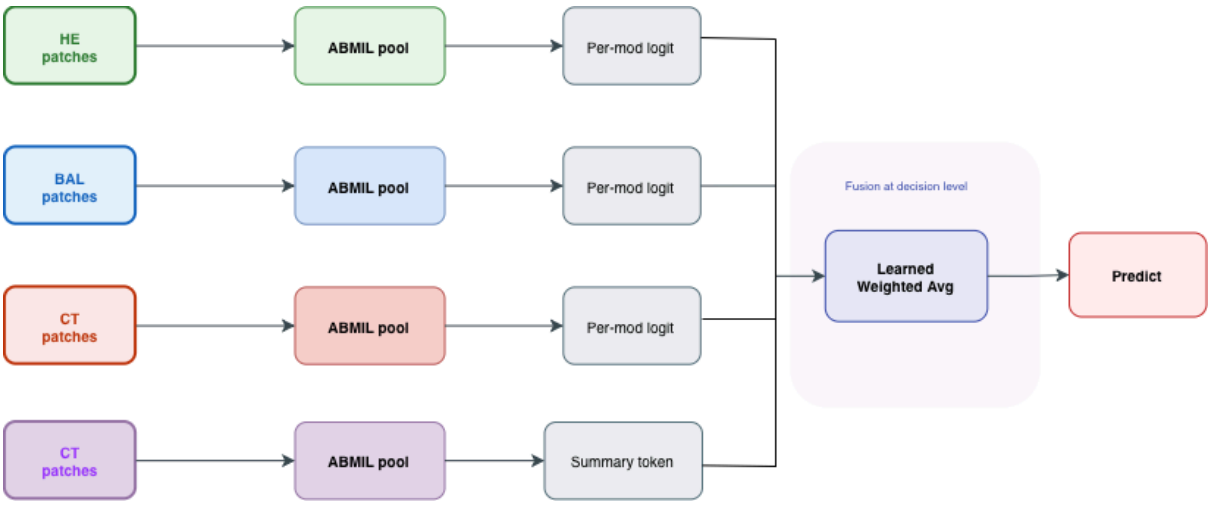
CLS Pooling: A learned CLS token is prepended to the token sequence. Cross-attention from the CLS token to the remaining tokens is computed via multi-head attention, followed by layer normalization. The CLS output is used as the bag representation.

5.2 Fusion Variants



We evaluate six distinct fusion strategies, each exploring different levels and mechanisms of cross-modal interaction:

5.2.1 Variant 1: Early Fusion (early / early_cls)

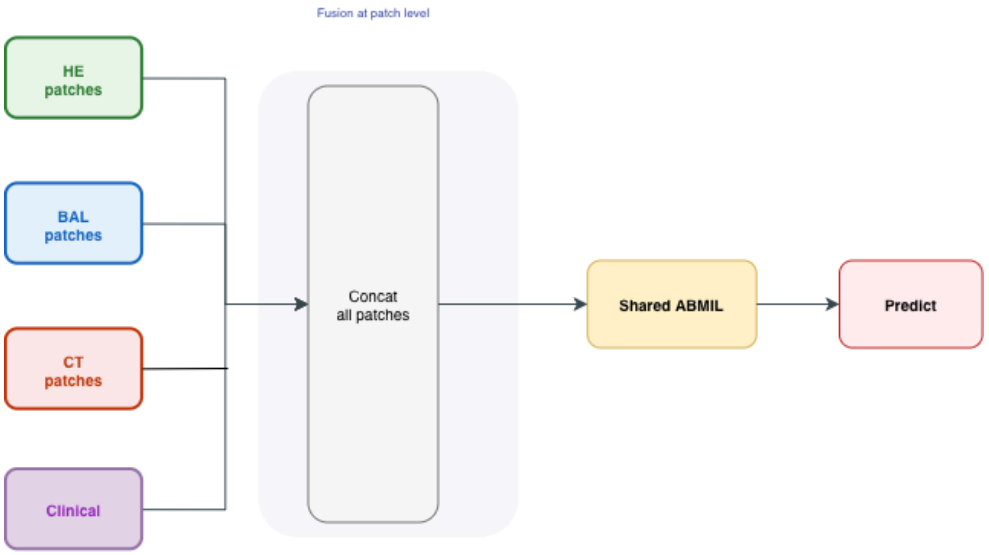


All modality patches are mapped to the shared feature space ℝ^H via their respective frozen backbones, concatenated into a single sequence, and pooled with ABMIL or a CLS token:

H_all = [H_HE ; H_BAL ; H_CT ; H_Clin] ∈ ℝ^((Σ□N□)×H)
ŷ = f_head(pool(H_all))

Modalities absent for a patient are simply omitted. Patches are sub-sampled to at most P_max = 2048 per modality.

5.2.2 Variant 2: Late Fusion (late)

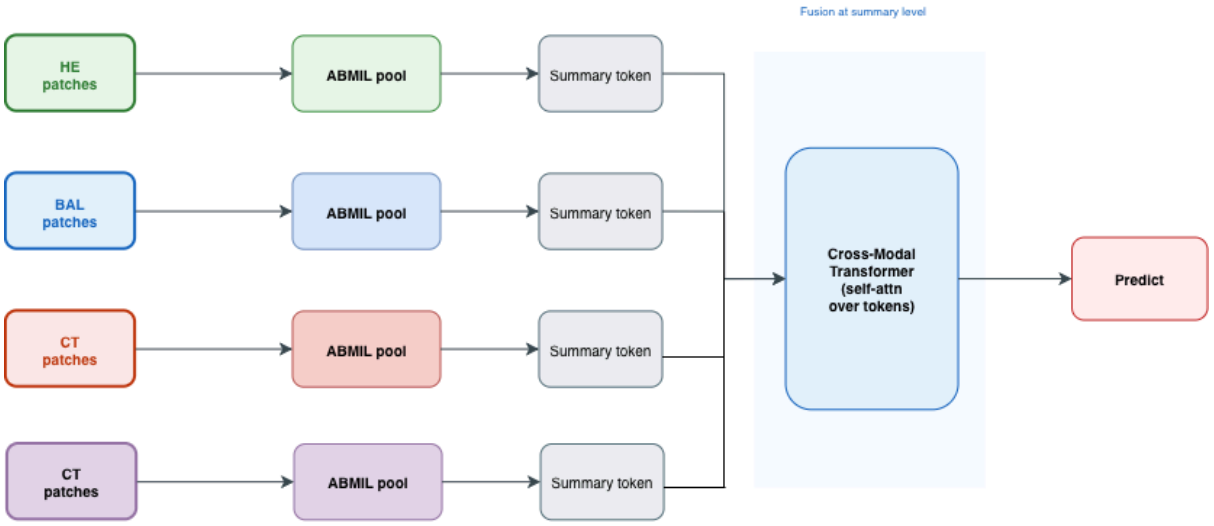


Each modality is processed independently through its ABMIL encoder to produce a logit $\hat{y}_m$. The final prediction is a softmax-weighted sum of present modality logits:

$$\hat{y} = \Sigma_{m \in M} w_m \hat{y}_m, \quad w = \text{softmax}(\ell[M])$$

where $\ell \in \mathbb{R}^{|M_0|}$ is a learned weight vector over the full modality set M_0, indexed by the present set M.

5.2.3 Variant 3: Middle Fusion (middle / middle_cls)

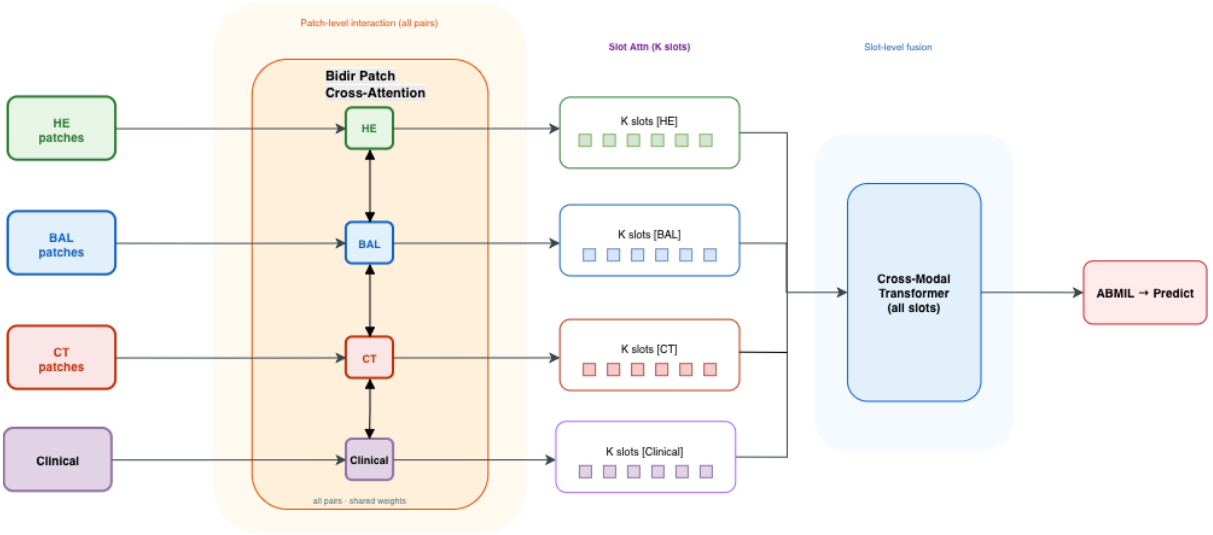


Each modality is reduced to a single summary vector by ABMIL. The |M| modality summaries are passed through the cross-modal transformer and pooled:

ŷ = f_head(pool(XFmr([r□]□∈M)))

The transformer is only applied when |M| ≥ 2.

5.2.4 Variant 4: Cross-Attention Fusion (crossattn / crossattn_cls)



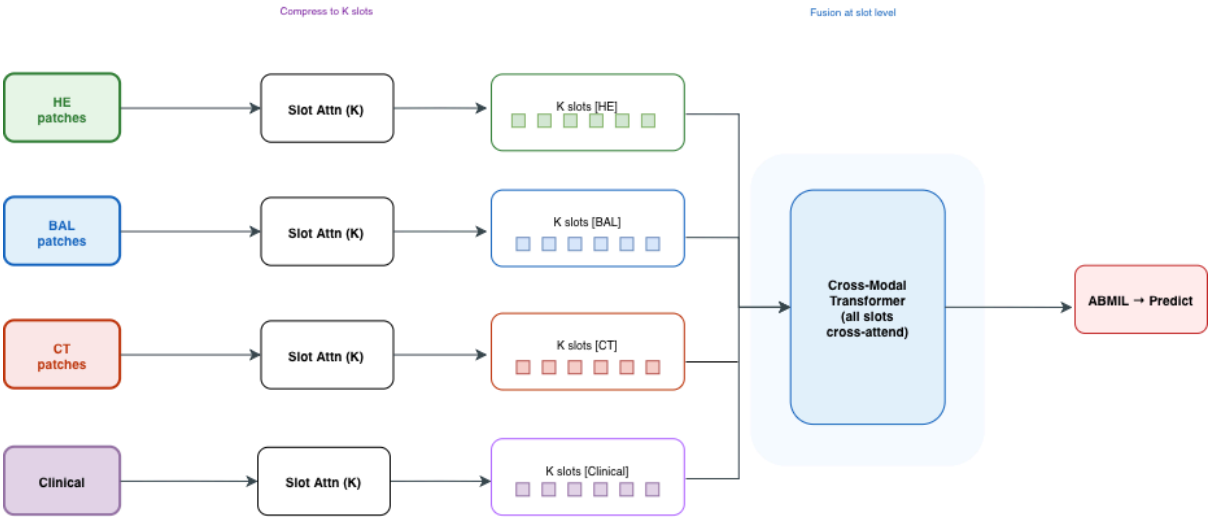
This variant first applies bidirectional patch-level cross-attention across modality pairs, then compresses each modality with Slot Attention. Cross-attention operates between modality patch sequences, not between individual patch instances across all modalities at once.

For each pair (m, m′), MCAT-style [2] bidirectional cross-attention produces cross-enriched patch features H̃□. Each enriched sequence is then summarized into K slot vectors by Slot Attention. The slot summaries from all modalities are concatenated and fused via the cross-modal transformer:

H̃□ = BidirCA(H□, H□′) ∀m′≠m
S□ = SlotAttn(H̃□) ∈ ℝ^(K×H)
ŷ = f_head(pool(XFmr([S□]□∈M)))

Patch sequences for bidirectional cross-attention are capped at 256 patches per modality to limit memory.

5.2.5 Variant 5: Cross-Modal Slot Fusion (crossmodal / crossmodal_cls)

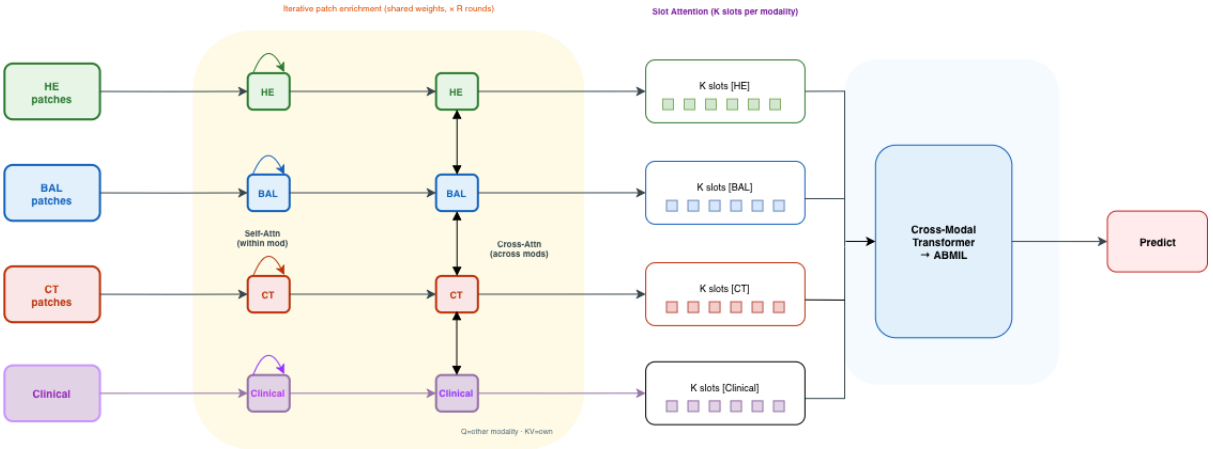


Each modality's patch sequence is first summarized into K slot vectors by Slot Attention. No patch-level cross-attention between modalities is performed. The slot summaries serve as compact modality representations. Cross-modal interaction then occurs at the level of these slot summaries through the cross-modal transformer:

$S_m = \text{SlotAttn}(H_m)$, $\hat{y} = f_{\text{head}}(\text{pool}(\text{XFmr}([S_m]_{m \in M})))$

All modalities share a single Slot Attention module but retain separate backbone encoders.

5.2.6 Variant 6: Iterative Cross-Modal Fusion (iterative / iterative_cls)



Inspired by Perceiver [8] and Multimodal Transformer [9], we apply R iterative blocks. Each block consists of within-modal self-attention followed by cross-modal cross-attention at the patch level. Let $H_m^{(r)}$ denote the patch features of modality m after block r.

Self-attention block:
$\tilde{H}_m^{(r)} = \text{FFN}(\text{LN}(H_m^{(r)}) + \text{MHA}(H_m^{(r)}, H_m^{(r)}, H_m^{(r)}))$

Cross-attention block:
$H^{(r+1)}\square = \mathrm{FFN}(\mathrm{LN}(\tilde{H}^{(r)}\square) + \mathrm{MHA}(\tilde{H}^{(r)}\square, C^{(r)}\text{-}\square, C^{(r)}\text{-}\square))$

where $C^{(r)}\text{-}\square = [\tilde{H}^{(r)}\square']\square'\neq\square$ is the concatenated patch context from all other modalities.

After R blocks, each modality is compressed via Slot Attention and fused with the cross-modal transformer identically to Variant 5:

$\hat{y} = f_head(pool(XFmr([SlotAttn(H^{(R)}\square)]\square \in M)))$

Default settings: R = 2, K ∈ {8, 16}. H&E patches are capped at 1024 inside the iterative blocks to control memory.

5.3 Phase 2 Summary Table

Table 4 provides a concise overview of the six fusion variants, highlighting the fusion level, key architectural components, and CLS token variant availability.

Variant	Fusion Level	Key Components	CLS Variant
early	patch	concat → pool	early_cls
late	logit	per-mod ABMIL, learned weight	—
middle	summary	ABMIL → XFmr → pool	middle_cls
crossattn	patch+slot	bidir-CA → SA → XFmr → pool	crossattn_cls
crossmodal	slot	SA → XFmr → pool	crossmodal_cls
iterative	patch+slot	R×(SA+CA) → SA → XFmr → pool	iterative_cls

Notation: CA = cross-attention, SA = slot attention, XFmr = cross-modal transformer, pool = ABMIL or CLS pooling

5.4 Phase 2 Hyperparameters

Table 5 summarizes the hyperparameters used during Phase 2 multimodal training:

Parameter	Symbol	Value
Learning rate	η_2	5×10^{-5}
Weight decay	—	1×10^{-3}
Max epochs	—	200
Eval frequency	—	Every 20 epochs
Gradient accumulation	—	4 steps
Transformer layers	L	2
Attention heads	nh	4
Attention dropout	—	0.1
Modal dropout	—	0.3
Max patches (total)	P_max	2048
Max HE (iterative)	—	1024
Slot iterations	T	3

6 Training Protocol

Optimizer

Adam [10] optimizer with learning rates and weight decay as specified in Tables 3 and 5. Mixed precision (FP16) training is employed via PyTorch AMP with gradient scaling. Gradient accumulation runs over 4 steps to simulate larger effective batch sizes.

Cross-Validation

Nested cross-validation strategy: 5 outer splits × 4 inner folds. Model selection is performed on the inner validation fold using balanced accuracy as the selection criterion.

Checkpoint Selection

For each fold, the checkpoint with the highest validation balanced accuracy is selected as the best model. This ensures robust model selection under class imbalance.

7 Cross-Attention Interpretability

7.1 ABMIL Attention Maps

The per-instance attention weights $\{a_i\}$ from the gated attention mechanism (Equation 4) provide a scalar importance score for every patch in each modality. Patches with high attention are those the model relies on most for its prediction.

For H&E modality, these patches can be projected back onto the biopsy slide using the stored tile coordinates (tile_left, tile_top), producing a spatial attention heatmap over the tissue section. This visualization reveals which histological regions drive the model's decision.

For clinical features, each row of the bag (one feature token) has an attention weight. The token label is bin-specific (e.g., 'FEV1_q3' or 'BMI_nan'), so the attention map reveals not only which clinical variables were attended but also whether the attended values were high or low.

7.2 Clinical × Cluster Cross-Attention

After Phase 2 inference, we compute an interpretability map that quantifies how strongly each clinical feature attends to each cell-cluster type.

Let F_Clin ∈ ℝ^(F×H) be the backbone projections of the F clinical-feature token rows, and let C□ ∈ ℝ^(K□×H) be the slot centroid tokens for modality m. All cluster tokens are concatenated:

C = [C□]□∈M ∈ ℝ^(Ktot×H)

The clinical-cluster cross-attention matrix is:

$$A_{clin} = \mathrm{softmax}(F_{Clin} C^{\top} / \sqrt{H}) \in [0,1]^{(F \times K_{tot})}$$

where the softmax is computed over K_{tot}.

Each clinical feature produces a probability distribution over all cluster tokens. Entry $A_{clin}[f, k]$ quantifies how much clinical feature f attends to cluster k. Each clinical feature token is labeled with its bin-specific identifier (e.g., 'FEV1_q3', 'BMI_nan'), so the attention map distinguishes which quantile bin of each feature is co-active with which cell population.

Cohort-level summaries are obtained by averaging A_{clin} over patients separately for rejection-positive and rejection-negative cases, then computing their difference to identify discriminative patterns.

8 Cluster-Level Interpretability

Each instance in a bag (patch, cell, or CT slice) carries a cluster-type annotation string (e.g., 'HE_cells/Lymphocytes') loaded from the pre-computed .pt file. These annotations enable five complementary cluster-level analyses that reveal which cell populations and clinical states the model attends to.

8.1 ABMIL Attention by Cluster Type

For each modality m, the ABMIL softmax attention weights $\alpha_m \in \mathbb{R}^{N_m}$ assign a scalar importance to every instance. Given cluster-type labels $c_i \in C_m$ for instance i, the mean attention for cluster type c is:

$$\bar{\alpha}_{m,c} = (1/|I_c|) \sum_{i \in I_c} \alpha_i, \quad I_c = \{i : c_i = c\}$$

where I_c is the index set of all instances of type c in the bag.

A bar chart of $\bar{\alpha}_{m,c}$ averaged over the cohort, shown separately for ACR-positive and ACR-negative patients, reveals which cell types are most attended by the ABMIL head in each clinical group. Differences between the two groups indicate cell types that are discriminative for rejection.

8.2 Modality Importance from Classifier Weights

For the late-fusion variant, each modality produces an independent logit and the final decision is a learned weighted sum:

$$\hat{y} = \sum_m w_m \hat{y}_m, \quad \{w_m\} = \mathrm{softmax}(\ell_{modal})$$

where ℓ_{modal} is a learnable parameter vector.

The weight w_m represents the classifier's estimate of how much modality m contributes to the final decision. Comparing the mean weights $\bar{w}^{+}_m$ (ACR) and $\bar{w}^{-}_m$ (non-ACR) reveals whether one modality dominates in each class.

8.3 Self-Attention Cluster Co-Attention

Within the iterative cross-modal model, each block applies a within-modality self-attention layer with weight matrix $W^{(r)} \in [0,1]^{(N \times N)}$ (patches as queries and keys). For the final block R, the cluster co-attention matrix $S \in \mathbb{R}^{(C \times C)}$ measures how strongly patches of type c_1 attend to patches of type c_2:

$$S[c_1, c_2] = (1/(|I_{c_1}||I_{c_2}|)) \sum_{i \in I_{c_1}} \sum_{j \in I_{c_2}} W^{(R)}[i, j]$$

where the patch sequence is capped at N_cap = 256 before aggregation.

This reveals intra-modality cell-cell attention patterns that differ between ACR and non-ACR patients.

8.4 Slot Attention Cluster Contribution

The slot attention module assigns each instance to K slots via a competition softmax. Let $A \in \mathbb{R}^{(K \times N)}$ be the final-round slot assignment matrix (post-softmax) for modality m. The K × C slot cluster contribution matrix is:

$$B[k, c] = (1/|I_c|) \sum_{i \in I_c} A[k, i]$$

giving the mean assignment weight that slot k places on instances of type c.

We compute both the post-softmax version (as shown) and the pre-softmax (raw logit) version, where $A^{pre_m}[k, i] = q_k^T k_i \cdot \text{scale}$ before the slot competition normalization. Side-by-side heatmaps reveal whether slot specialization is already present in the raw scores or emerges only after competition.

8.5 Cross-Modal Slot Connection

For slot-based fusion variants, the cross-modal transformer operates on the concatenated slot sequence $Z = [Z_1; \ldots; Z_M]$, yielding a full attention matrix $X \in \mathbb{R}^{(K_{tot} \times K_{tot})}$. We extract the inter-modality block $X_{ab} \in \mathbb{R}^{(K_a \times K_b)}$ for modality pair (m_a, mb).

Three-Panel Connection Figure

For a given modality pair, we display:

1. Left: $B_a \in \mathbb{R}^{(K_a \times C_a)}$, showing which cell types load onto each m_a slot
2. Center: $X_{ab} \in \mathbb{R}^{(K_a \times K_b)}$, showing cross-modal attention between m_a and mb slot summaries
3. Right: $B^T_b \in \mathbb{R}^{(C_b \times K_b)}$, showing which mb cell types each mb slot represents

Reading left to right traces the information path: m_a cell type → m_a slot → cross-modal → mb slot → mb cell type.

Cluster-to-Cluster Cross-Modal Attention

To summarize the multi-hop path into a single Ca × Cb matrix, we compute:

$Pab = B^{T}\square_a Xab\ B\square b \in \mathbb{R}^{\wedge}(Ca\times Cb)$

where entry $Pab[c_1, c_2]$ quantifies the effective attention from cell type c_1 in m_a to cell type c_2 in mb.

Heatmaps of Pab for ACR vs. non-ACR, and their difference, reveal which inter-modality cell-type interactions are specific to rejection.

ACR vs. Non-ACR Comparison

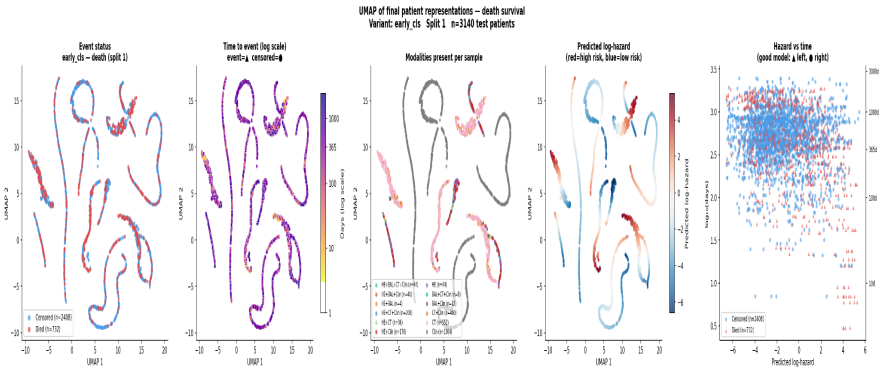
All cluster-level interpretability plots are shown in three panels: mean over ACR-positive patients, mean over ACR-negative patients, and the signed difference Δ = μ_ACR − μ_non-ACR. Positive entries in Δ indicate mechanisms that are more active in rejection. Negative entries indicate mechanisms that are more active in non-rejection.

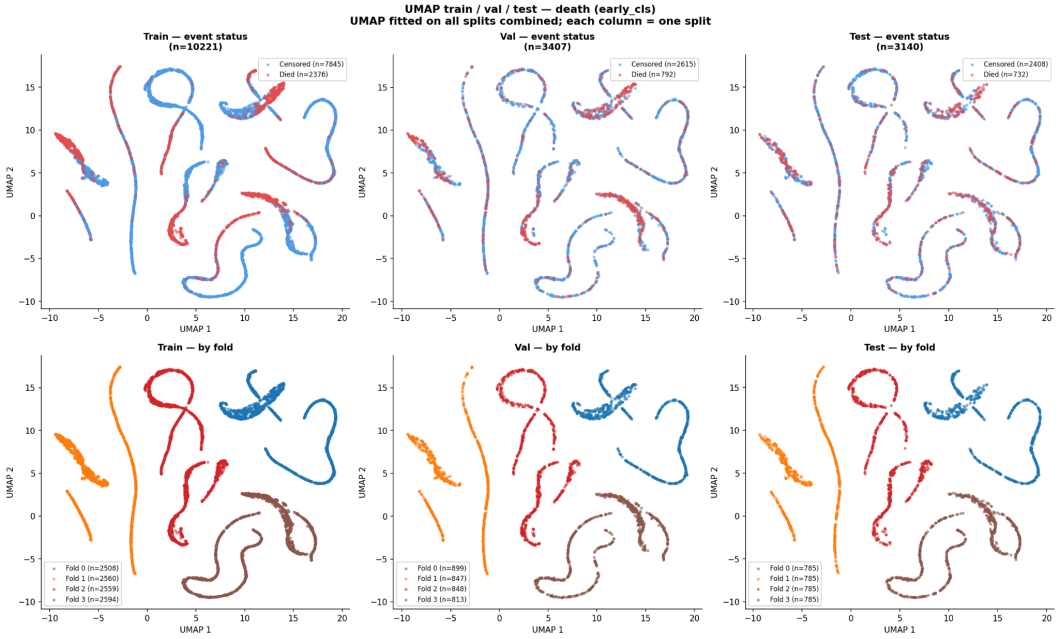
9 Results

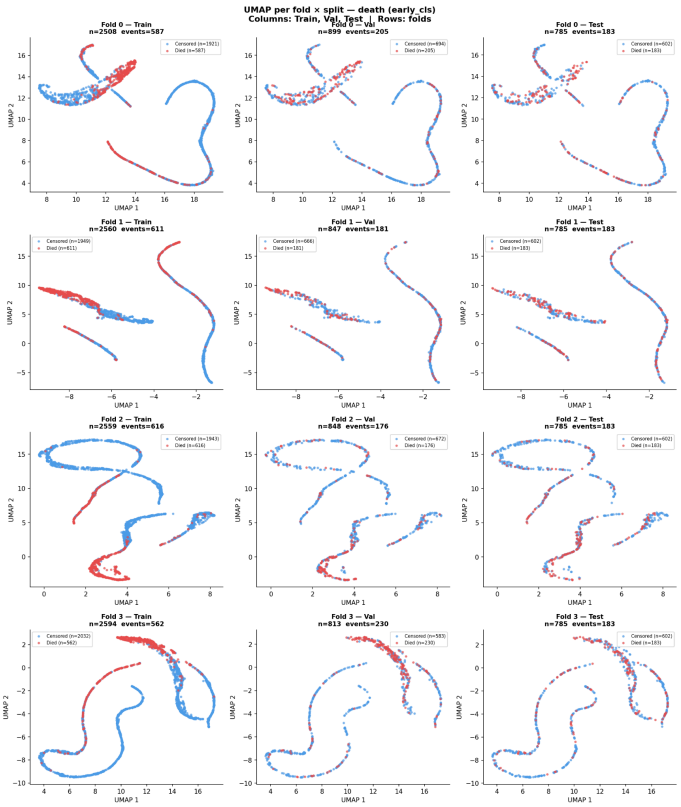
9.1 Death Survival Prediction

The death prediction task asks the model to rank patients by their time-to-death risk: a higher predicted risk score should correspond to shorter observed survival. This is evaluated using the Concordance Index (C-index), which measures the fraction of all comparable patient pairs for which the model correctly identifies the one who died sooner.

UMAP Visualisation of Embedding Space







The UMAP plots (Figure: ‘UMAP train/val/test – death’) show the 2D projection of the early_cls model’s learned patient representations across training, validation, and test splits. Points are coloured by survival status (censored vs. died).

Key observations from the UMAP:

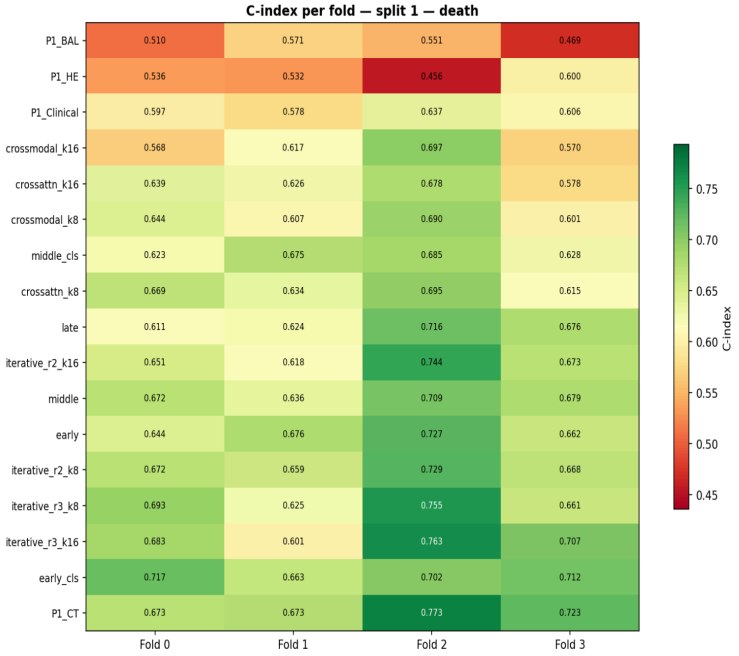
- The embedding space forms elongated, curved manifolds rather than tight clusters. This is characteristic of a continuous risk score: patients at different points along the manifold correspond to different levels of survival risk, rather than hard class boundaries.

- Deceased patients (red dots) are not perfectly separated from censored patients (blue dots), which reflects the inherent difficulty of the task – many patients who are currently censored will eventually die, and the model predicts relative risk rather than binary outcome.

- The consistency of structure across train, validation, and test splits (bottom row, coloured by fold) indicates that the model is learning a stable representation that generalises across patient subsets rather than overfitting to specific folds.

- The spread of deceased patients along the manifold suggests the model captures a gradient of risk, consistent with a well-calibrated survival model.

C-index Performance Across Fusion Variants



The heatmap ('C-index per fold – split 1 – death') shows the test C-index for each fusion variant across 4 folds. Several important patterns emerge:

Unimodal baselines: The unimodal models (P1_BAL, P1_HE, P1_Clinical, P1_CT) establish a performance floor. P1_CT achieves the highest unimodal C-index (mean ~0.67), reflecting the strong prognostic signal in CT imaging for long-term outcomes. P1_BAL is the weakest unimodal predictor, consistent with BAL's primary role as a rejection marker rather than a mortality predictor. P1_Clinical and P1_HE show moderate performance (~0.58–0.60).

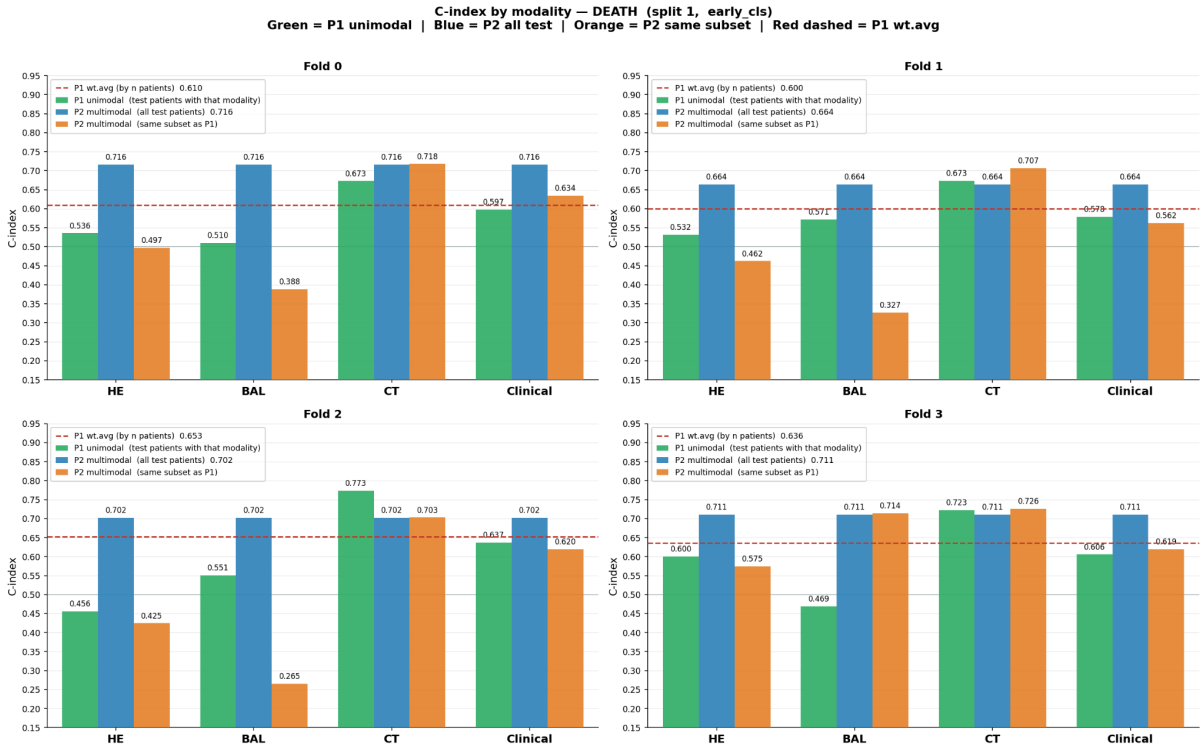
Multimodal models: Most multimodal variants outperform the unimodal baselines, demonstrating the benefit of integrating complementary modalities. The early_cls variant

achieves the highest mean C-index (~0.71–0.72 across folds), suggesting that simple concatenation with CLS pooling is surprisingly competitive. Iterative variants (iterative_r3_k8, iterative_r3_k16) also perform strongly (~0.68–0.76), indicating that iterative cross-modal interaction at the patch level is beneficial.

Fold variability: The C-index shows moderate fold-to-fold variability, particularly for BAL and some multimodal variants in Fold 2. This is expected given the limited sample size for some modality combinations and the inherent difficulty of survival prediction. The variability highlights the importance of cross-validation for reliable performance estimation.

Unimodal vs. Multimodal Comparison (Bar Charts)

The per-fold bar charts compare: (green) Phase 1 unimodal performance on the subset of test patients that have that specific modality, (blue) Phase 2 multimodal performance on all test patients, and (orange) Phase 2 multimodal performance on the same subset as P1.



Key findings:

- CT is consistently the strongest individual modality, with unimodal C-indices of 0.71–0.77 across folds. This is the most informative single modality for death prediction.

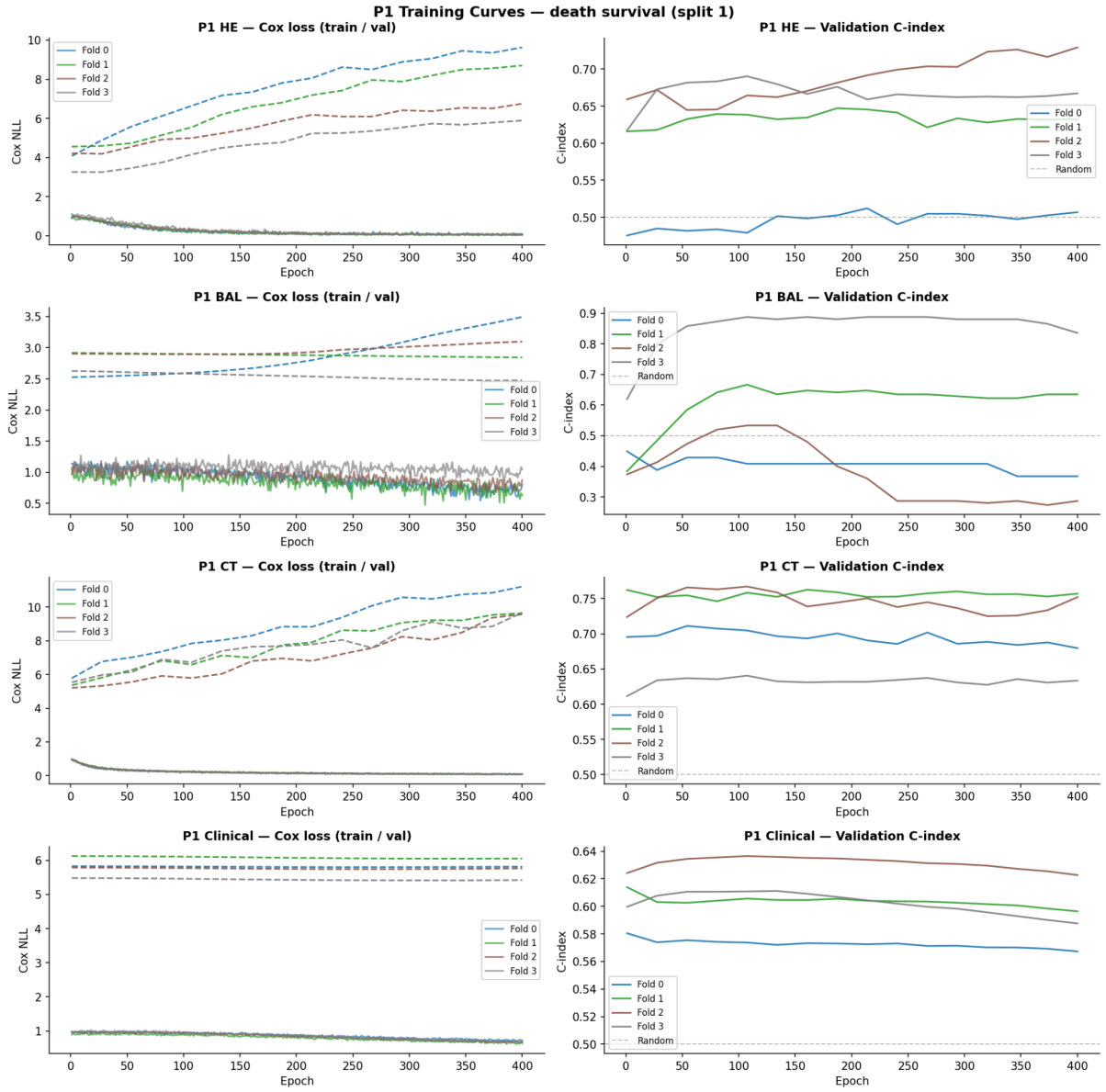
- The multimodal model (blue bars) generally meets or exceeds the weighted average unimodal performance (P1 wt. avg., shown as dashed lines), confirming that fusion adds value over any single modality.

- The orange bars (P2 on the P1 subset) are sometimes lower than the green bars (P1 on the same subset). This is expected: the multimodal model is trained and evaluated on the full patient population (including patients with only one modality), while the unimodal model was specialised for patients with that modality. The multimodal model trades some modality-specific accuracy for breadth of coverage.

- BAL shows high variability (C-index ranging from ~0.25 to ~0.85 across folds), likely due to the small number of patients with BAL data available at each timepoint.

Phase 1 Training Curves

The training curves (P1 Training Curves – death survival) show the Cox NLL loss (train/val) and validation C-index over 400 epochs for each modality and fold.



Key observations:

- HE encoder: Training loss decreases monotonically, but validation loss increases after ~100 epochs (classic overfitting pattern). The validation C-index stabilises at 0.60–0.70, indicating the model learns useful representations but is limited by the relatively small number of H&E samples with linked survival labels.

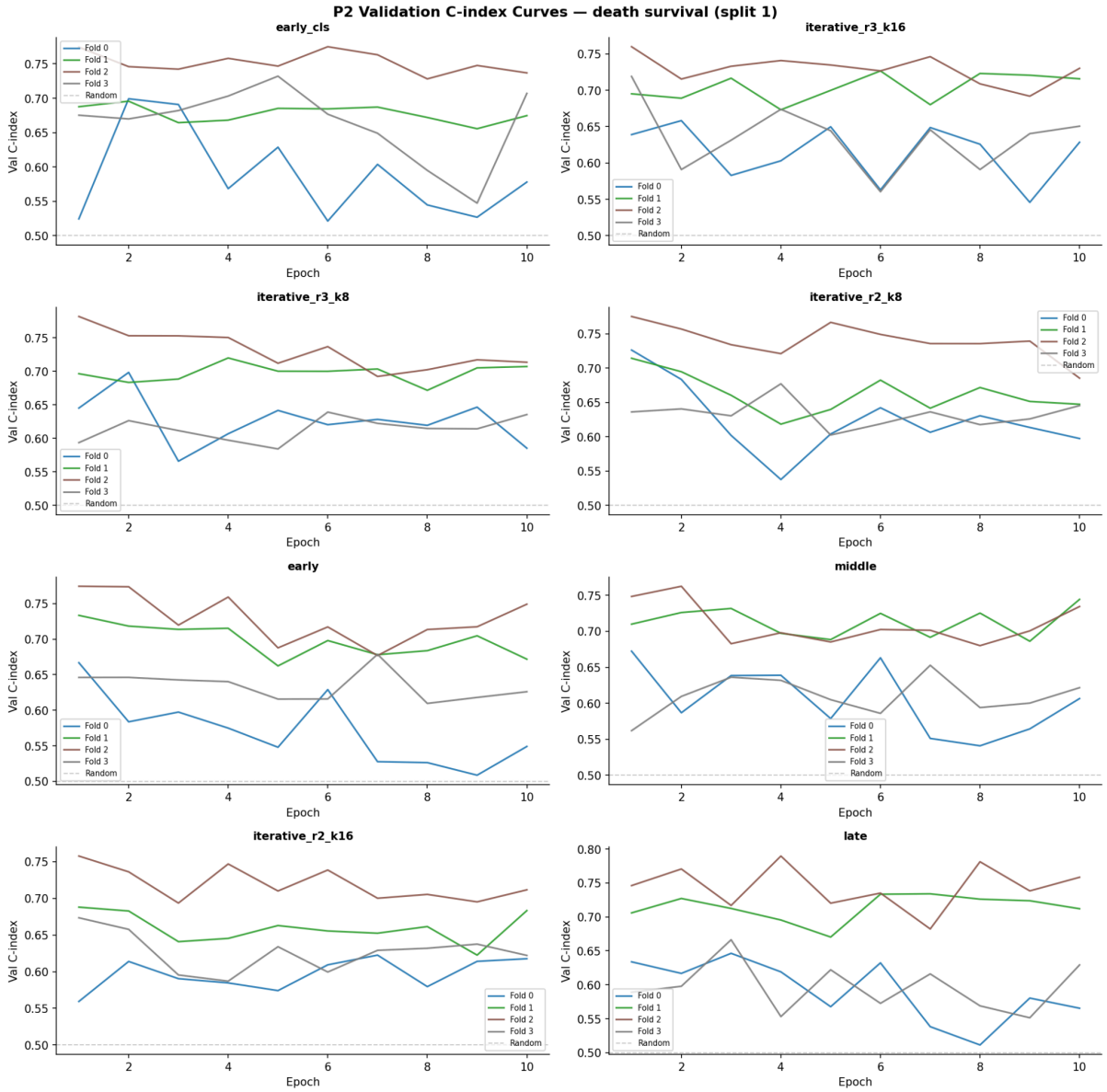
- BAL encoder: Both training and validation losses are low and stable (Cox NLL ~1.0), and the validation C-index is highly variable (0.3–0.9). The low dimensionality of BAL features (10-dimensional scVI embeddings) limits the encoder's capacity.

- CT encoder: The most stable training curves. Validation C-index plateaus at 0.65–0.75 across folds, consistent with CT being the strongest unimodal predictor. The gap between training and validation loss is smaller than for HE, suggesting less overfitting.

- Clinical encoder: Very stable training (Cox NLL ≈1.0 throughout), with validation C-index of 0.57–0.62. The clinical encoder learns quickly and plateaus early, reflecting that clinical features are relatively low-variance after tokenisation.

Phase 2 Validation C-index Curves

The Phase 2 validation curves show C-index over 10 evaluation epochs for the main fusion variants.



Key observations:

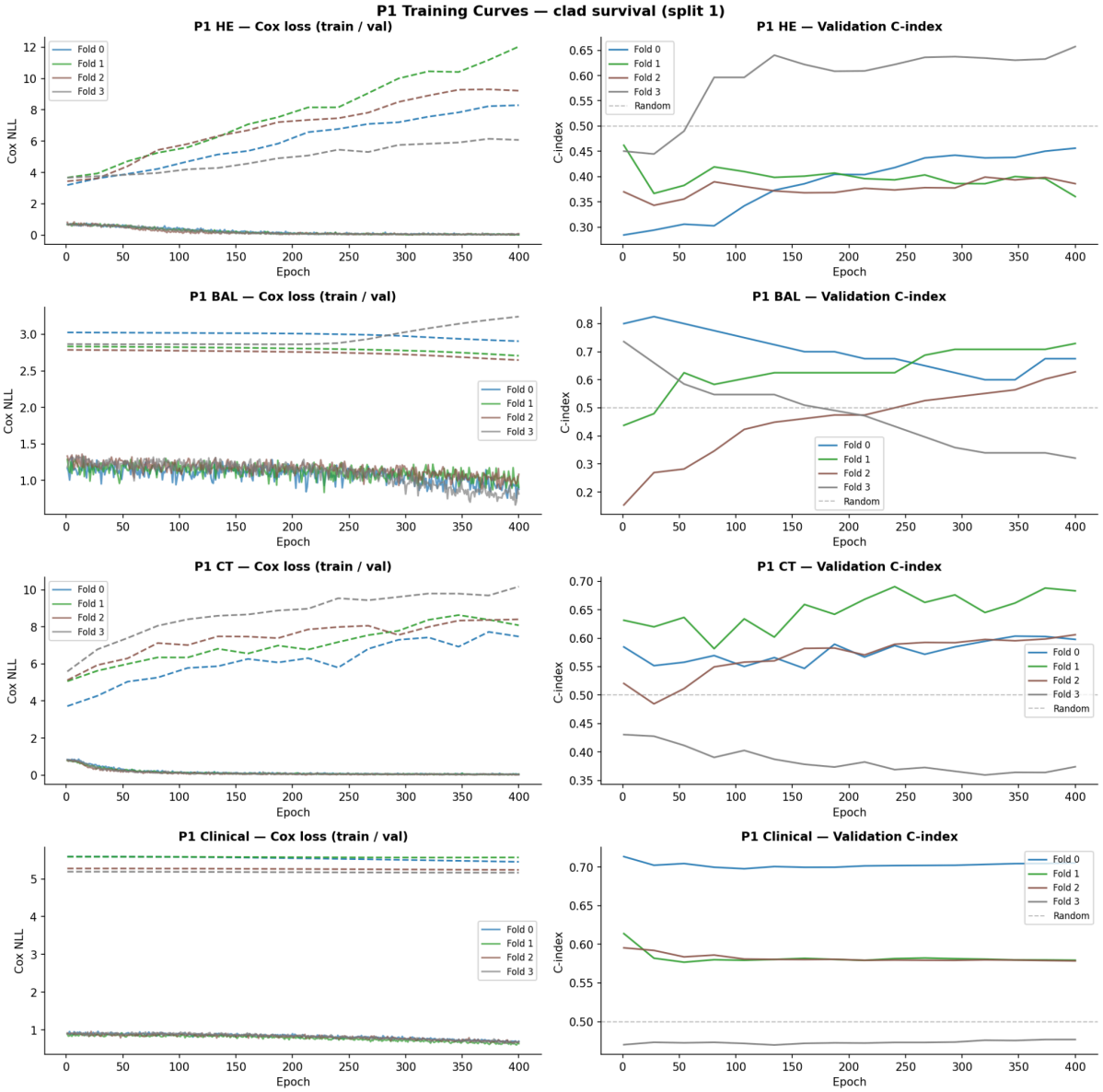
- early_cls achieves the highest peak validation C-index (~0.75) in Fold 0–1 and converges quickly (by epoch 4–6). The CLS pooling mechanism provides a strong inductive bias for this task.

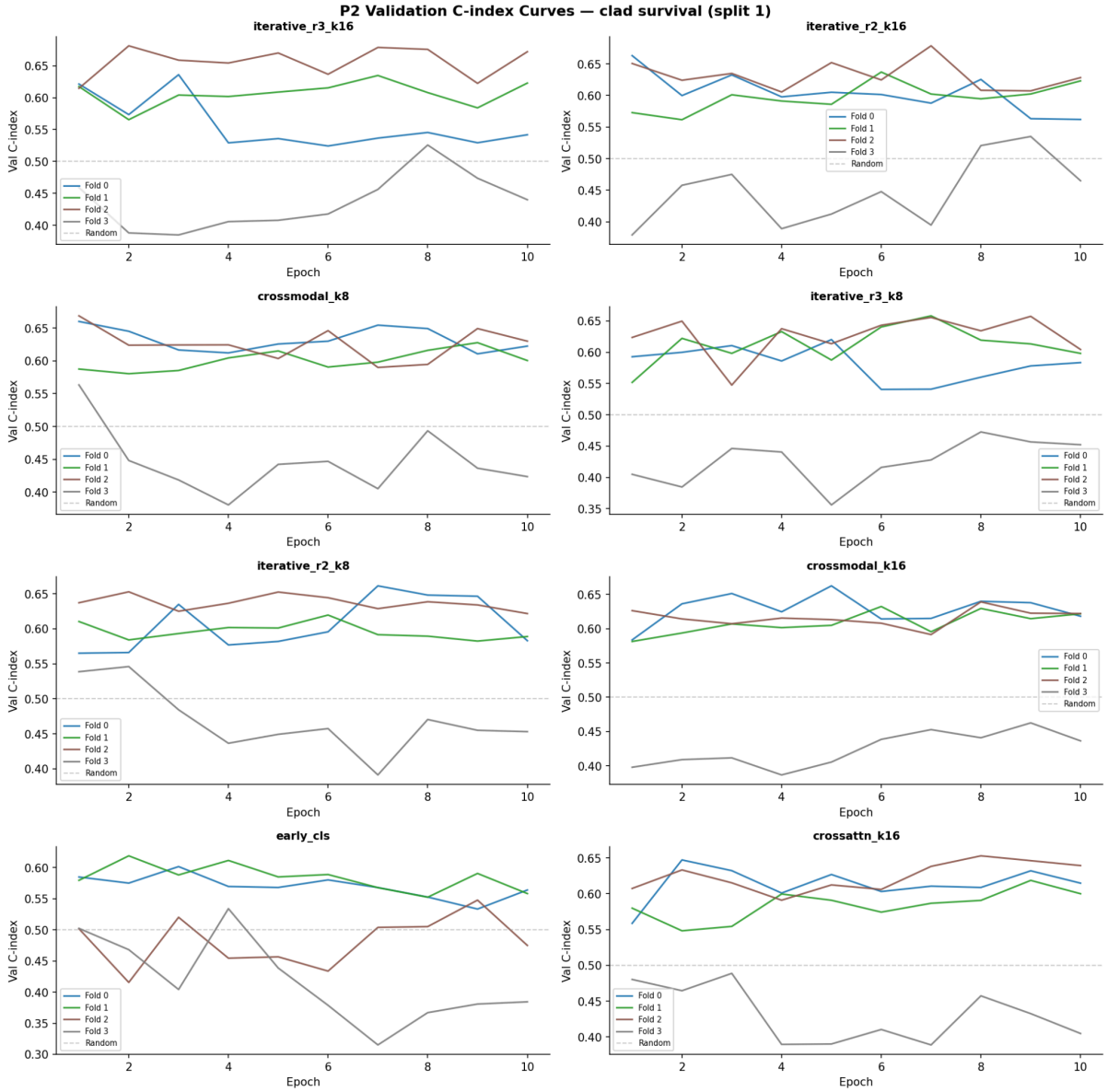
- Iterative variants (iterative_r2_k8, iterative_r2_k16, iterative_r3_k8, iterative_r3_k16) show comparable peak performance to early_cls, with slightly more variability across folds. Increasing the number of iterations (r2 vs. r3) and slots (k8 vs. k16) produces modest improvements.

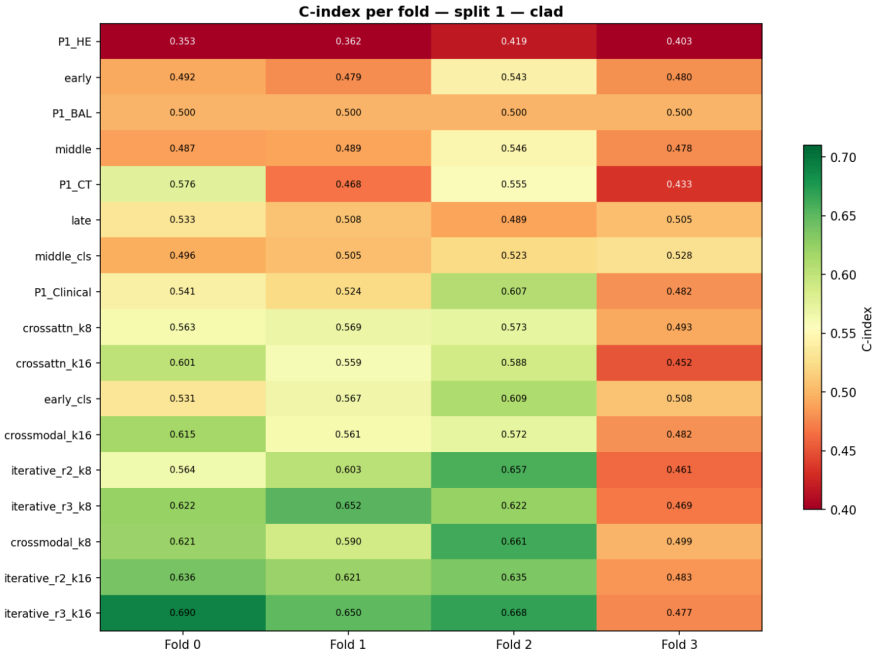
- The late fusion variant achieves C-indices of ~0.70–0.77 with relatively stable convergence, suggesting that modality-specific logits combined with learned weights are a strong baseline.

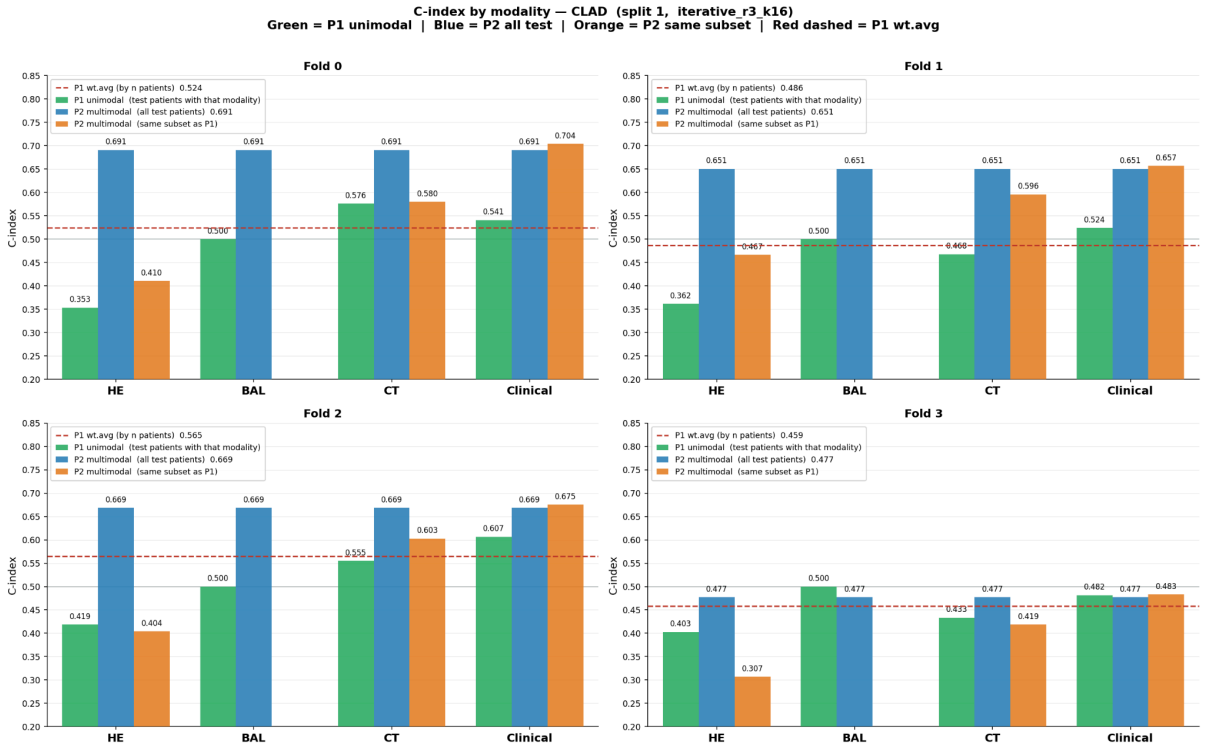
- All variants substantially exceed the random baseline (C-index = 0.5, shown as the grey dashed line), confirming that the model learns meaningful survival-discriminative representations.

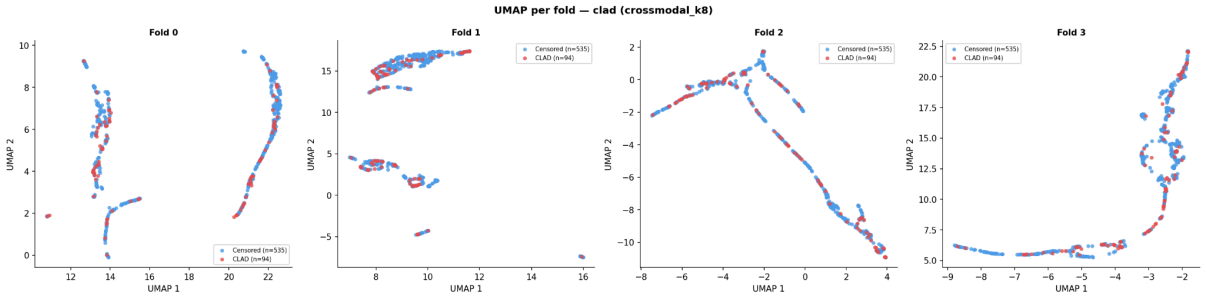
9.2 CLAD Occurrence Prediction

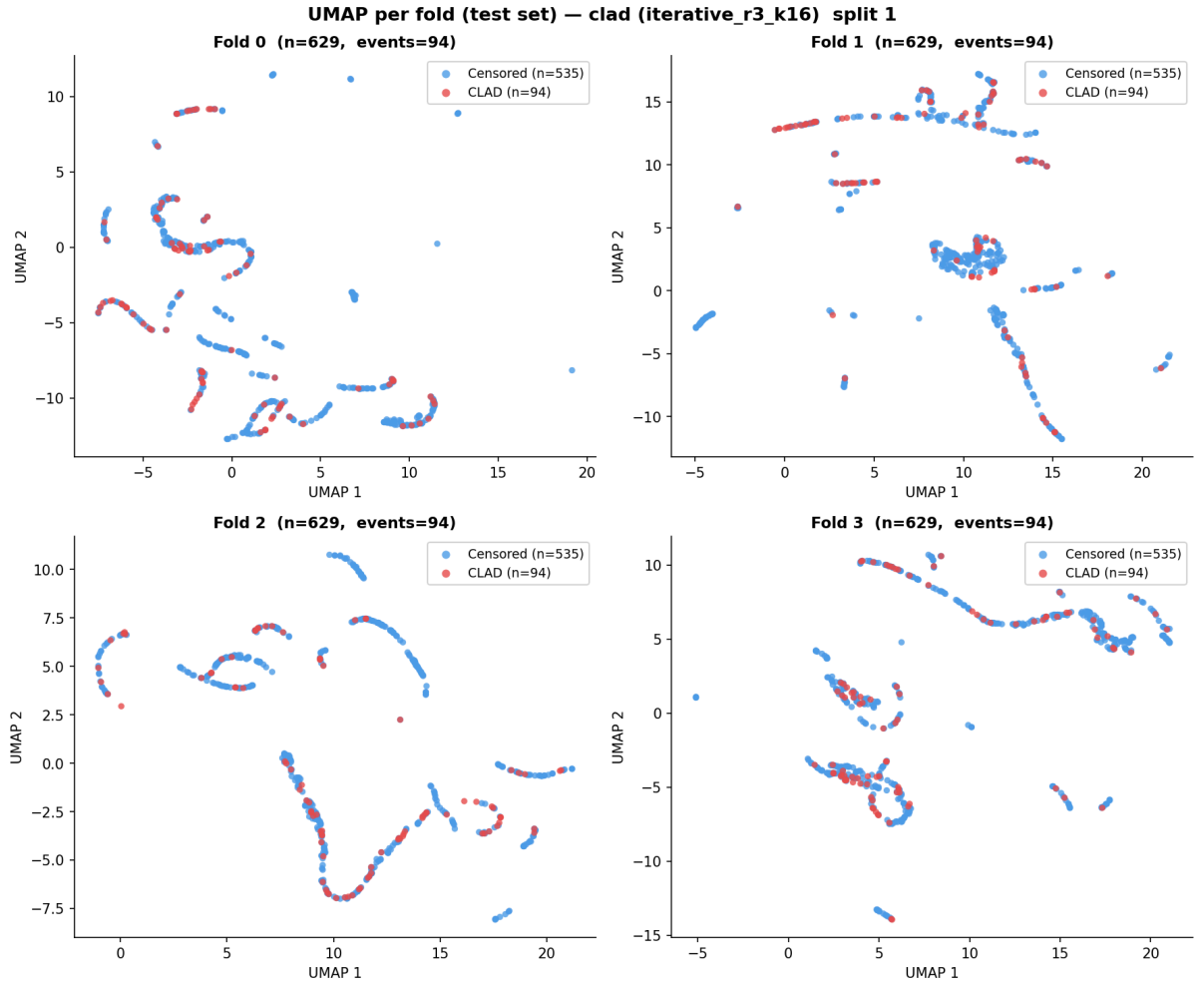

P1 Training Curves — clad survival (split 1)
P1 HE — Cox loss (train / val)
P1 HE — Validation C-index
P1 BAL — Cox loss (train / val)
P1 BAL — Validation C-index
P1 CT — Cox loss (train / val)
P1 CT — Validation C-index
P1 Clinical — Cox loss (train / val)
P1 Clinical — Validation C-index
Cox NLL
C-index
Epoch
Fold 0
Fold 1
Fold 2
Fold 3
Random

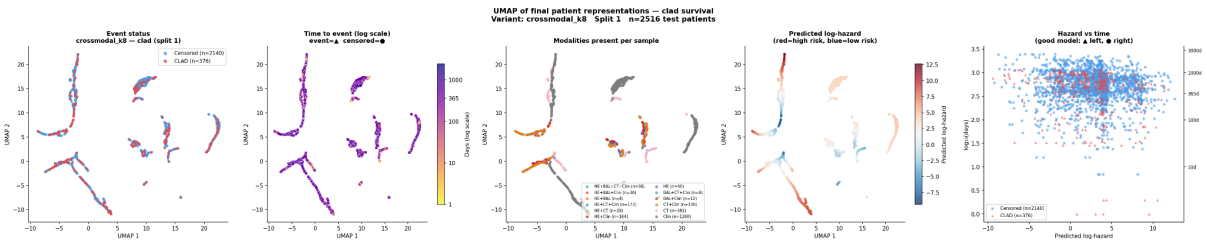

P2 Validation C-index Curves — clad survival (split 1)
iterative_r3_k16
iterative_r2_k16
crossmodal_k8
iterative_r3_k8
iterative_r2_k8
crossmodal_k16
early_cls
crossattn_k16
Val C-index
Epoch
Fold 0
Fold 1
Fold 2
Fold 3
Random

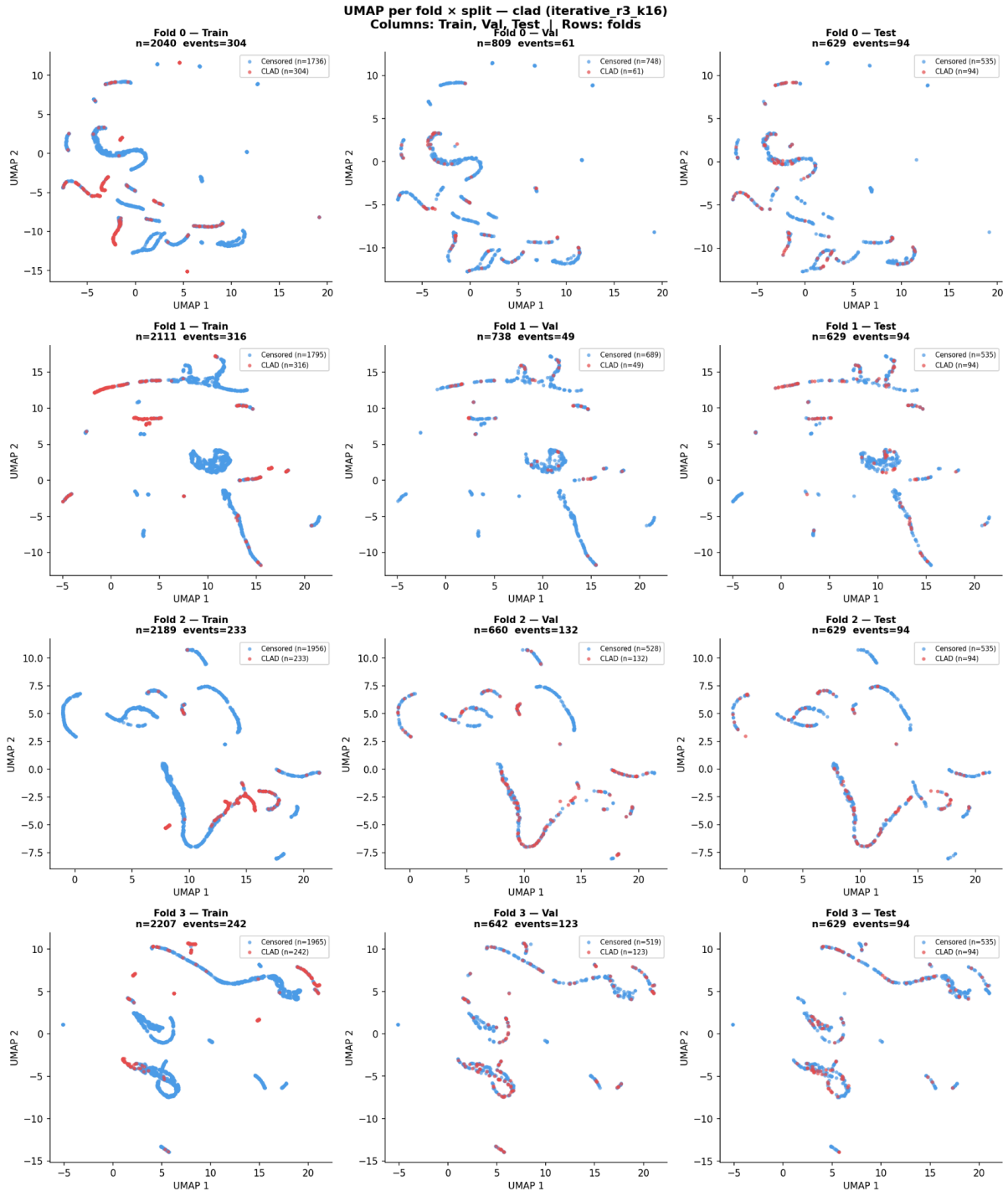

C-index per fold — split 1 — clad
P1_HE 0.353 0.362 0.419 0.403
early 0.492 0.479 0.543 0.480
P1_BAL 0.500 0.500 0.500 0.500
middle 0.487 0.489 0.546 0.478
P1_CT 0.576 0.468 0.555 0.433
late 0.533 0.508 0.489 0.505
middle_cls 0.496 0.505 0.523 0.528
P1_Clinical 0.541 0.524 0.607 0.482
crossattn_k8 0.563 0.569 0.573 0.493
crossattn_k16 0.601 0.559 0.588 0.452
early_cls 0.531 0.567 0.609 0.508
crossmodal_k16 0.615 0.561 0.572 0.482
iterative_r2_k8 0.564 0.603 0.657 0.461
iterative_r3_k8 0.622 0.652 0.622 0.469
crossmodal_k8 0.621 0.590 0.661 0.499
iterative_r2_k16 0.636 0.621 0.635 0.483
iterative_r3_k16 0.690 0.650 0.668 0.477
Fold 0
Fold 1
Fold 2
Fold 3
C-index
0.70
0.65
0.60
0.55
0.50
0.45
0.40

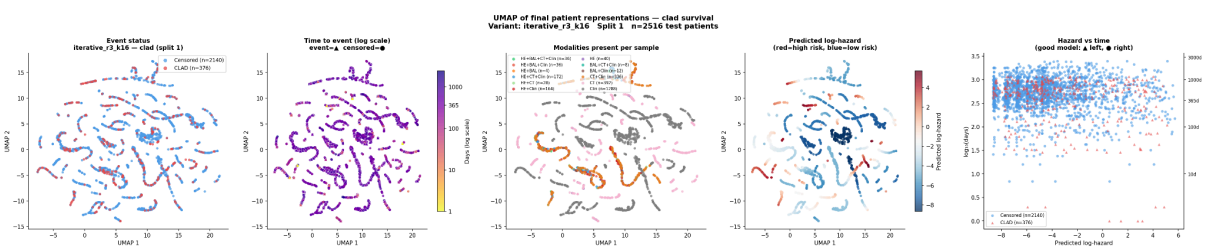

C-index by modality — CLAD (split 1, iterative_r3_k16)
Green = P1 unimodal | Blue = P2 all test | Orange = P2 same subset | Red dashed = P1 wt.avg
Fold 0
P1 wt.avg (by n patients) 0.524
P1 unimodal (test patients with that modality)
P2 multimodal (all test patients) 0.691
P2 multimodal (same subset as P1)
0.353 0.691 0.410
0.500 0.691
0.576 0.691 0.580
0.541 0.691 0.704
Fold 1
P1 wt.avg (by n patients) 0.486
P1 unimodal (test patients with that modality)
P2 multimodal (all test patients) 0.651
P2 multimodal (same subset as P1)
0.362 0.651 0.467
0.500 0.651
0.468 0.651 0.596
0.524 0.651 0.657
Fold 2
P1 wt.avg (by n patients) 0.565
P1 unimodal (test patients with that modality)
P2 multimodal (all test patients) 0.669
P2 multimodal (same subset as P1)
0.419 0.669 0.404
0.500 0.669
0.555 0.669 0.603
0.607 0.669 0.675
Fold 3
P1 wt.avg (by n patients) 0.459
P1 unimodal (test patients with that modality)
P2 multimodal (all test patients) 0.477
P2 multimodal (same subset as P1)
0.403 0.477 0.307
0.500 0.477
0.433 0.477 0.419
0.482 0.477 0.483
C-index
HE
BAL
CT
Clinical


UMAP per fold — clad (crossmodal_k8)
Fold 0
Fold 1
Fold 2
Fold 3
Censored (n=535)
CLAD (n=94)
UMAP 1
UMAP 2

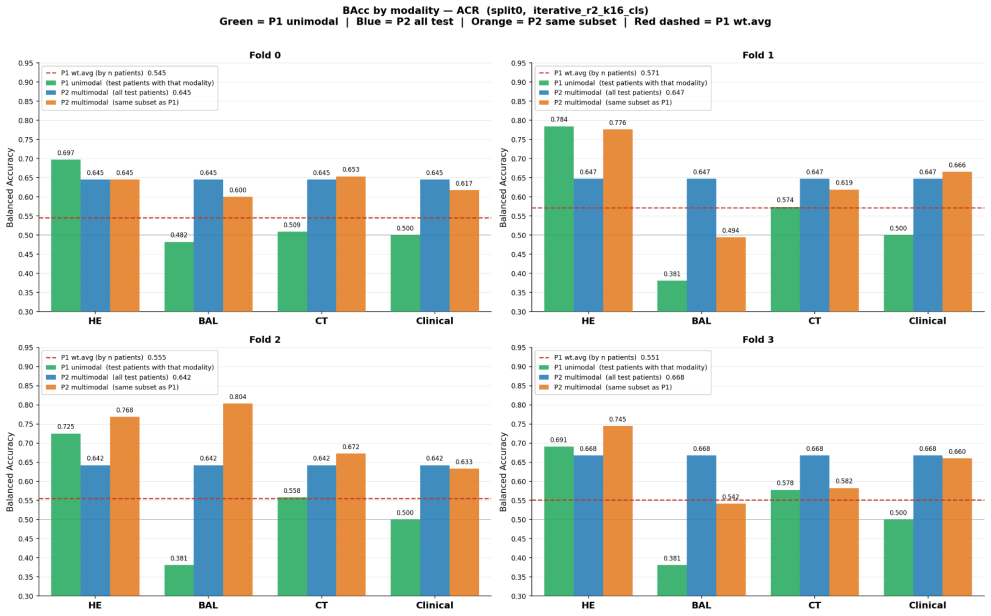

UMAP per fold (test set) — clad (iterative_r3_k16) split 1
Fold 0 (n=629, events=94)
Fold 1 (n=629, events=94)
Fold 2 (n=629, events=94)
Fold 3 (n=629, events=94)
Censored (n=535)
CLAD (n=94)
UMAP 1
UMAP 2

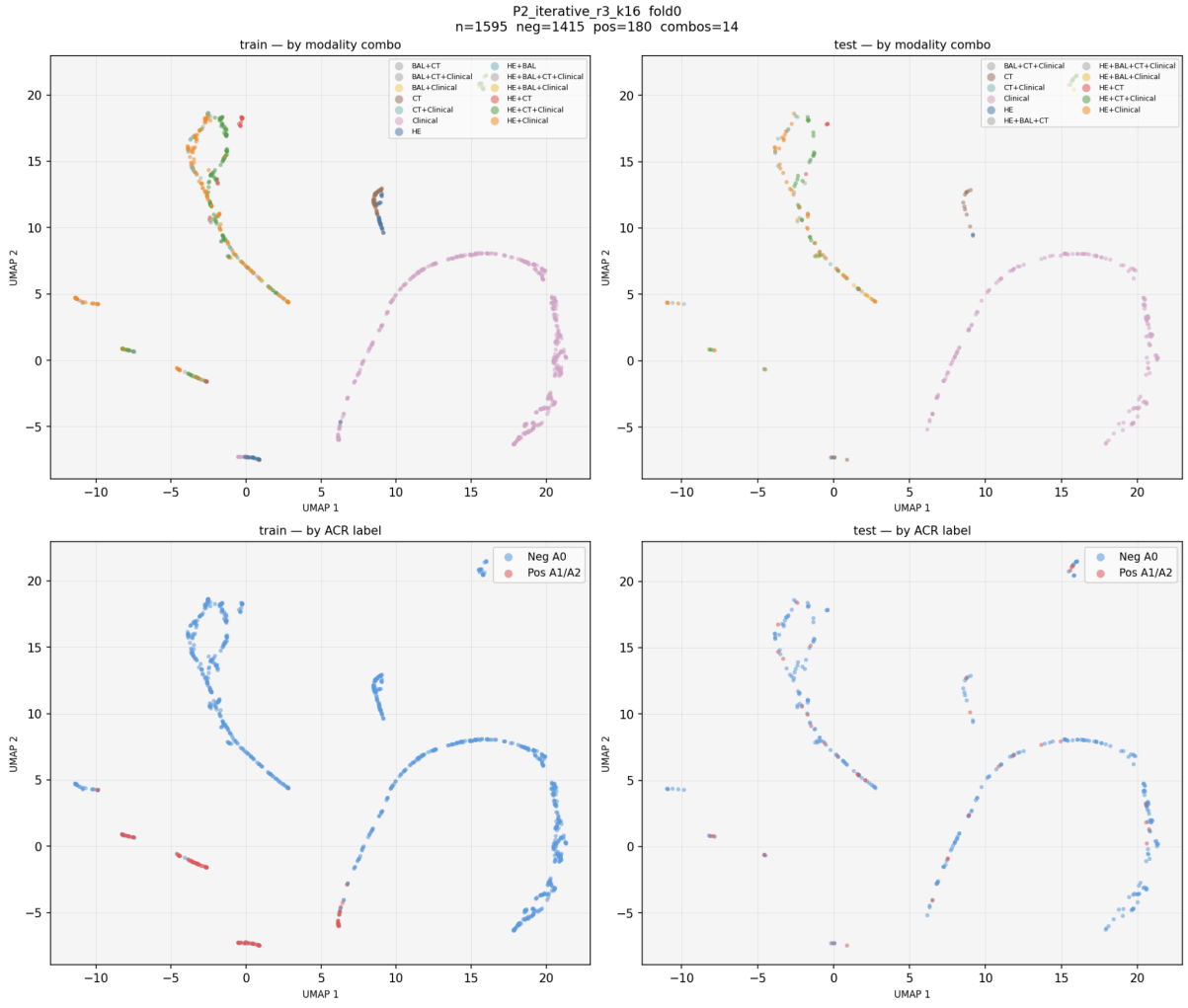

UMAP of final patient representations — clad survival
Variant: crossmodal_k8 Split 1 n=2516 test patients
Event status
crossmodal_k8 — clad (split 1)
Time to event (log scale)
event=▲ censored=●
Modalities present per sample
Predicted log-hazard
(red=high risk, blue=low risk)
Hazard vs time
(good model: ▲ left, ● right)
UMAP 1
UMAP 2
Predicted log-hazard

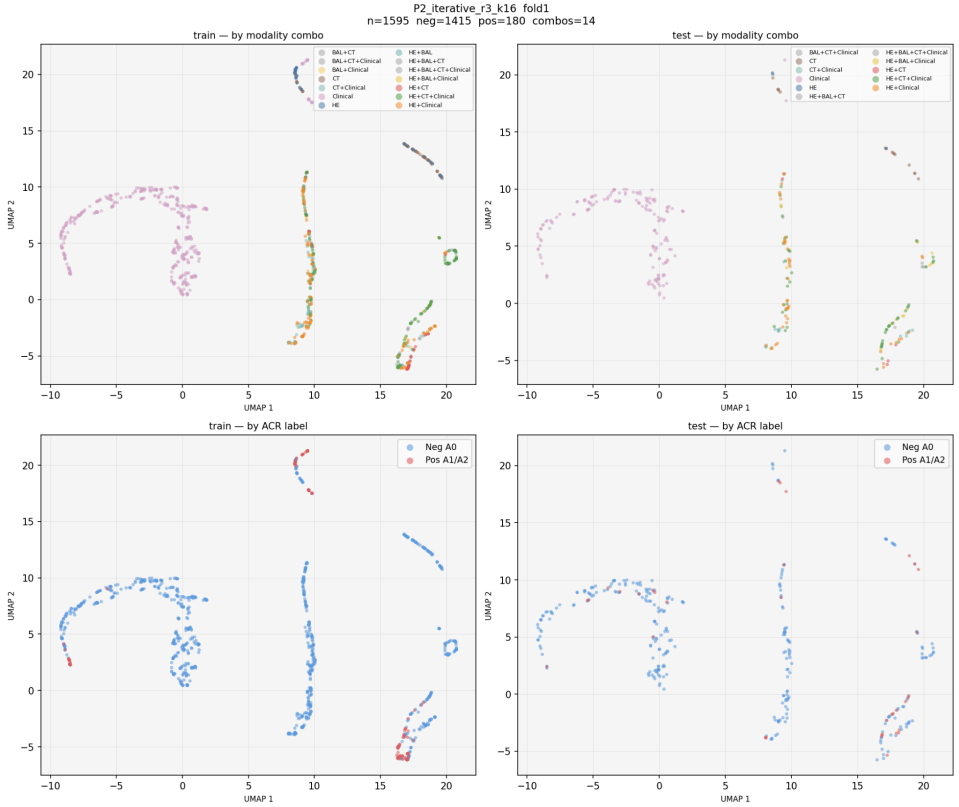

UMAP per fold × split — clad (iterative_r3_k16)
Columns: Train, Val, Test | Rows: folds
Fold 0 — Train
n=2040 events=304
Censored (n=1736)
CLAD (n=304)
Fold 0 — Val
n=809 events=61
Censored (n=748)
CLAD (n=61)
Fold 0 — Test
n=629 events=94
Censored (n=535)
CLAD (n=94)
Fold 1 — Train
n=2111 events=316
Censored (n=1795)
CLAD (n=316)
Fold 1 — Val
n=738 events=49
Censored (n=689)
CLAD (n=49)
Fold 1 — Test
n=629 events=94
Censored (n=535)
CLAD (n=94)
Fold 2 — Train
n=2189 events=233
Censored (n=1956)
CLAD (n=233)
Fold 2 — Val
n=660 events=132
Censored (n=528)
CLAD (n=132)
Fold 2 — Test
n=629 events=94
Censored (n=535)
CLAD (n=94)
Fold 3 — Train
n=2207 events=242
Censored (n=1965)
CLAD (n=242)
Fold 3 — Val
n=642 events=123
Censored (n=519)
CLAD (n=123)
Fold 3 — Test
n=629 events=94
Censored (n=535)
CLAD (n=94)
UMAP 1
UMAP 2

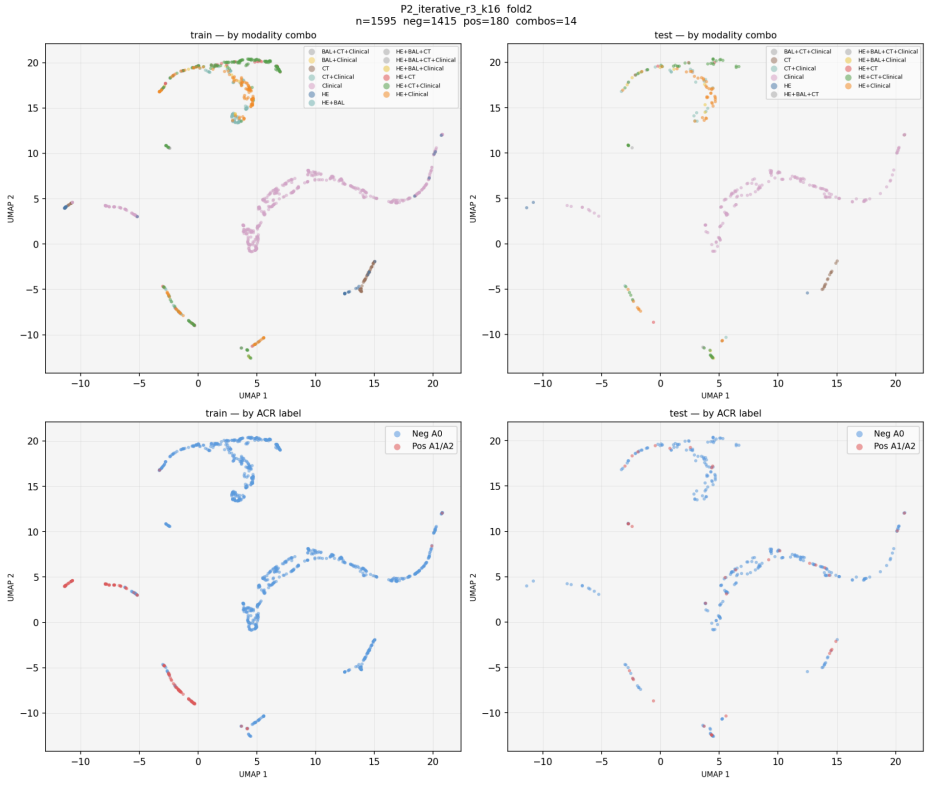

UMAP of final patient representations — clad survival
Variant: iterative_r3_k16 Split 1 n=2516 test patients
Event status
iterative_r3_k16 — clad (split 1)
Time to event (log scale)
event=▲ censored=●
Modalities present per sample
Predicted log-hazard
(red=high risk, blue=low risk)
Hazard vs time
(good model: ▲ left, ● right)
UMAP 1
UMAP 2
Predicted log-hazard

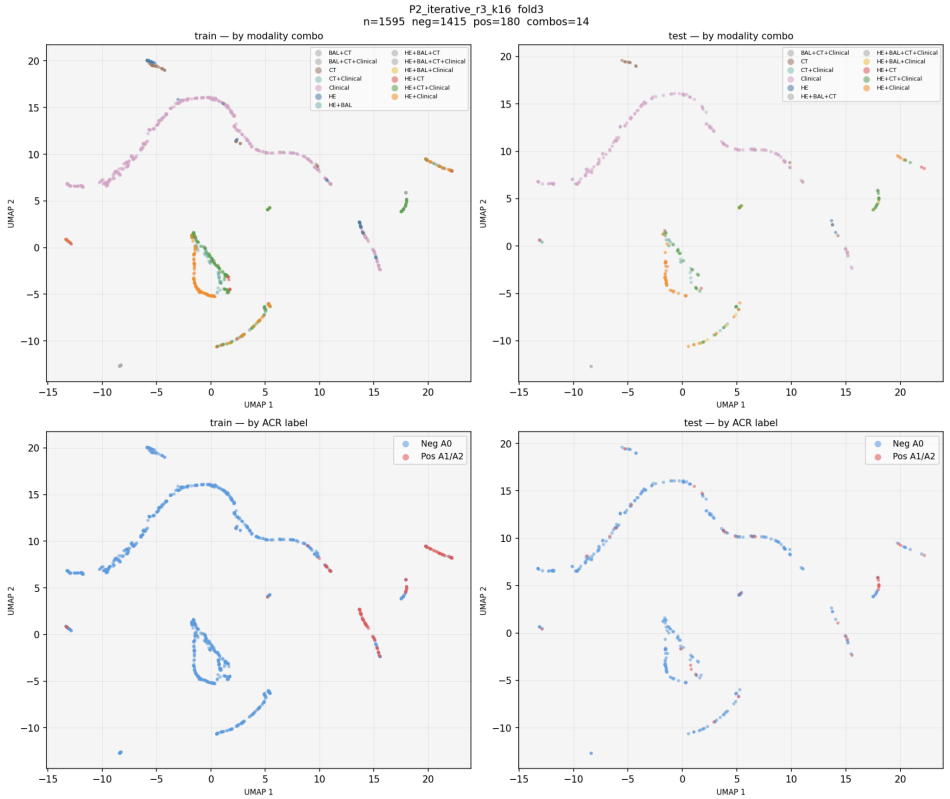
9.3 ACR classification






P2_iterative_r3_k16 fold1
n=1595 neg=1415 pos=180 combos=14
train — by modality combo
test — by modality combo
train — by ACR label
test — by ACR label
BAL+CT
BAL+CT+Clinical
BAL+Clinical
CT
CT+Clinical
Clinical
HE
HE+BAL
HE+BAL+CT
HE+BAL+CT+Clinical
HE+BAL+Clinical
HE+CT
HE+CT+Clinical
HE+Clinical
Neg A0
Pos A1/A2
UMAP 1
UMAP 2


P2_iterative_r3_k16 fold2
n=1595 neg=1415 pos=180 combos=14
train — by modality combo
test — by modality combo
train — by ACR label
test — by ACR label
Neg A0
Pos A1/A2
UMAP 1
UMAP 2


P2_iterative_r3_k16 fold3
n=1595 neg=1415 pos=180 combos=14
train — by modality combo
test — by modality combo
train — by ACR label
test — by ACR label
Neg A0
Pos A1/A2
UMAP 1
UMAP 2

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